



Signal and Waveforms: Signal

Signal and System

- Signals are variables that carry information
- **System** is an assemblage of entities/objects, real or abstract, comprising a whole with each every component/element interacting or related to another one.

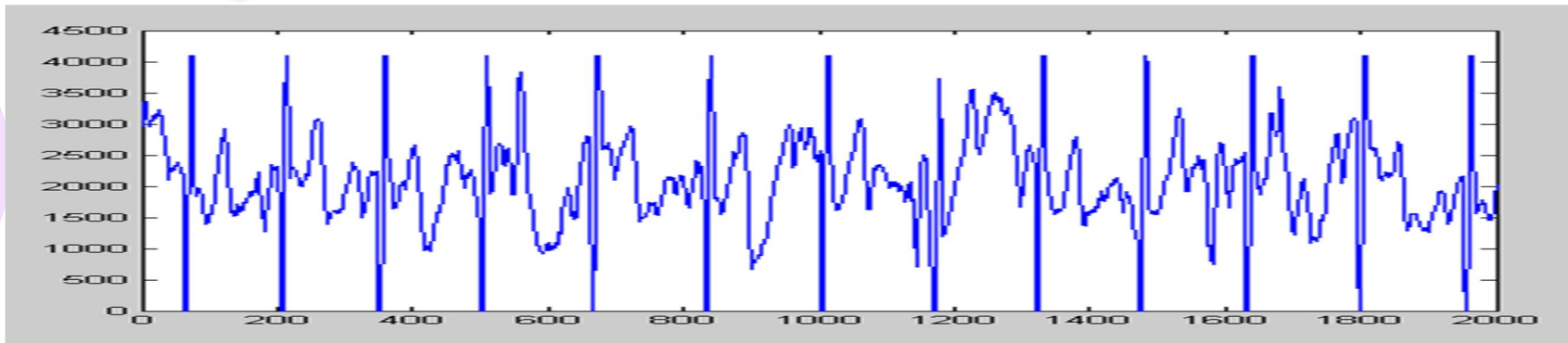
Systems process input signals to produce output signals



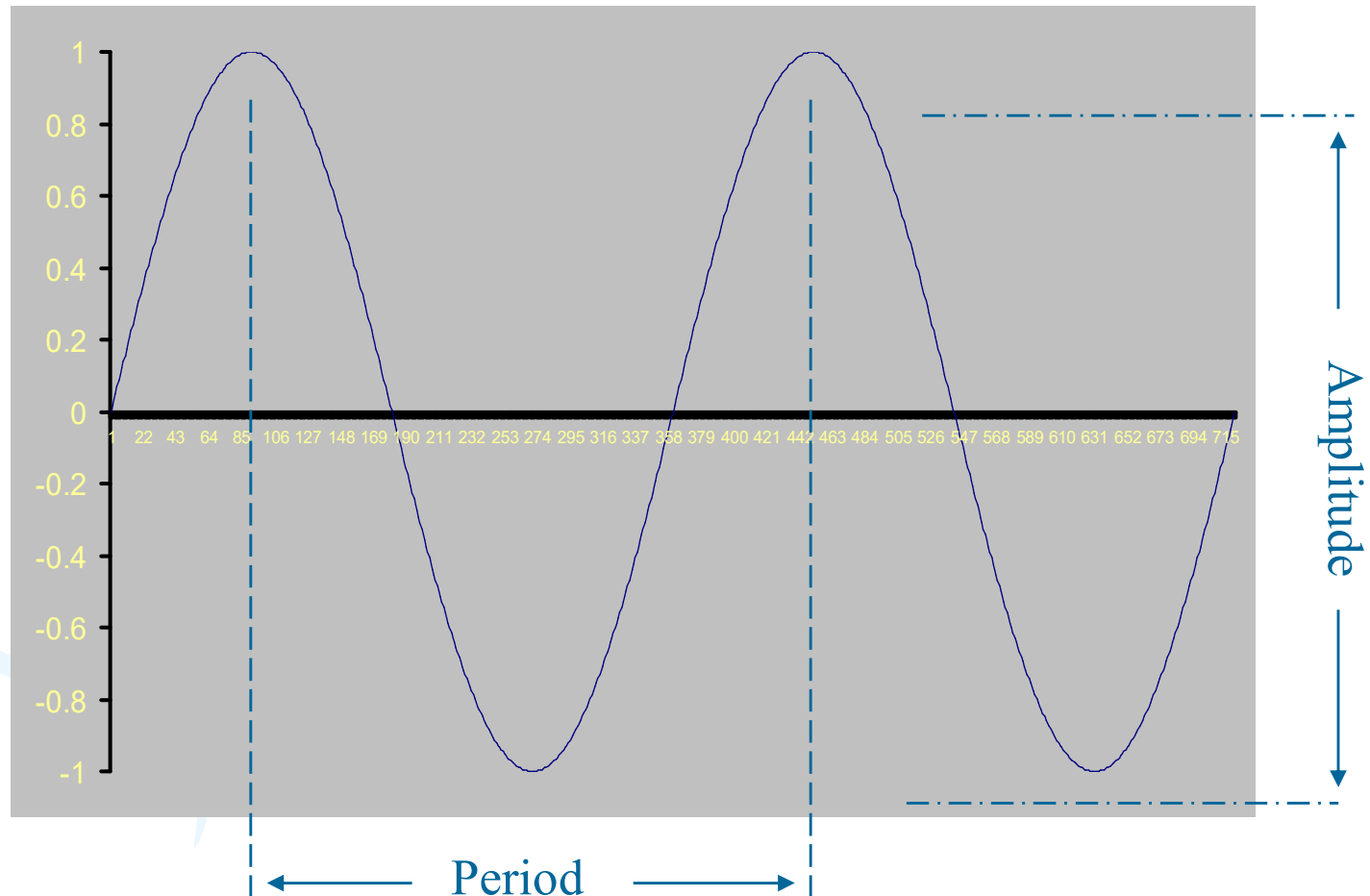
- Examples
 - i. Motion, sound, picture, video, traffic light...
 - ii. Natural system (ecosystem), human-made system (machines, computer storage system), abstract system (traffic, computer programs), descriptive system (plans)

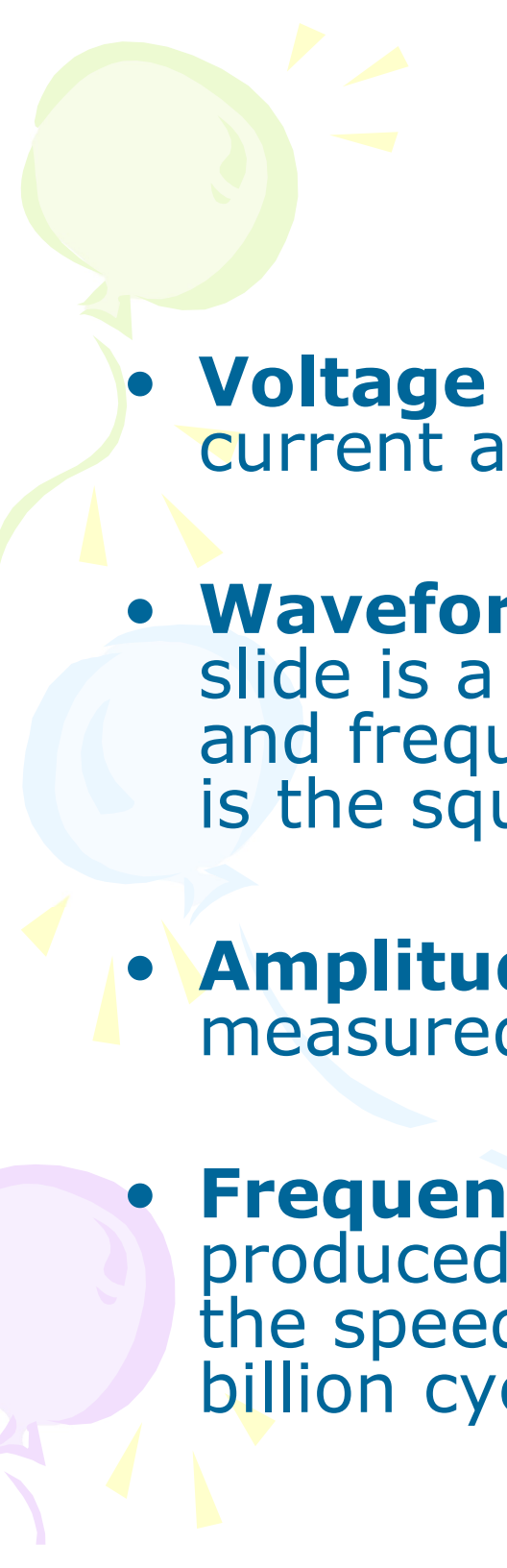
Signal Examples

- Electrical signals --- voltages and currents in a circuit
- Acoustic signals --- audio or speech signals (analog or digital)
- Video signals --- intensity variations in an image (e.g. a CAT scan)
- Biological signals --- sequence of bases in a gene
- Noise: unwanted signal



Measuring Signals





Definitions

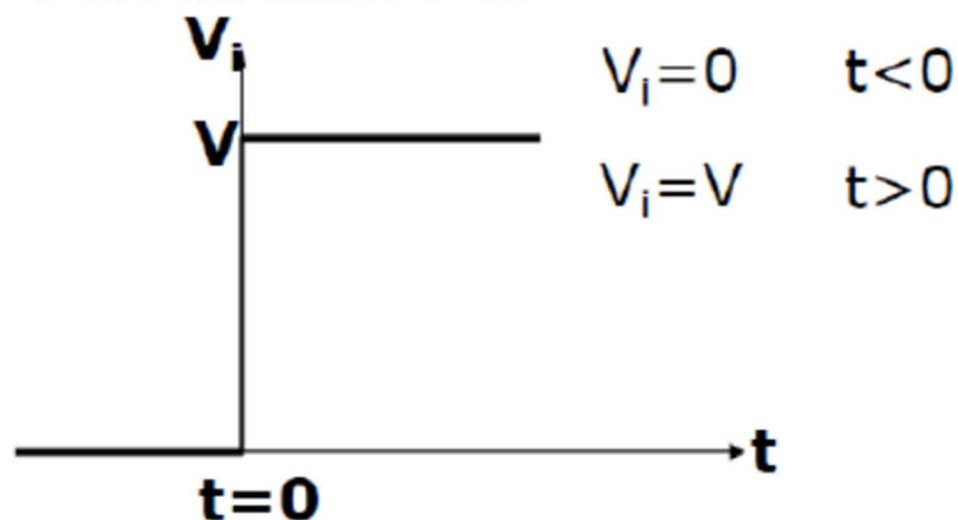
- **Voltage** – the force which moves an electrical current against resistance
- **Waveform** – the shape of the signal (previous slide is a sine wave) derived from its amplitude and frequency over a fixed time (other waveform is the square wave)
- **Amplitude** – the maximum value of a signal, measured from its average state
- **Frequency** (pitch) – the number of cycles produced in a second – Hertz (Hz). Relate this to the speed of a processor eg 1.4GigaHertz or 1.4 billion cycles per second

Non-sinusoidal waveforms

- 1) Step
- 2) Pulse
- 3) Square wave
- 4) Ramp
- 5) Exponential waveforms.

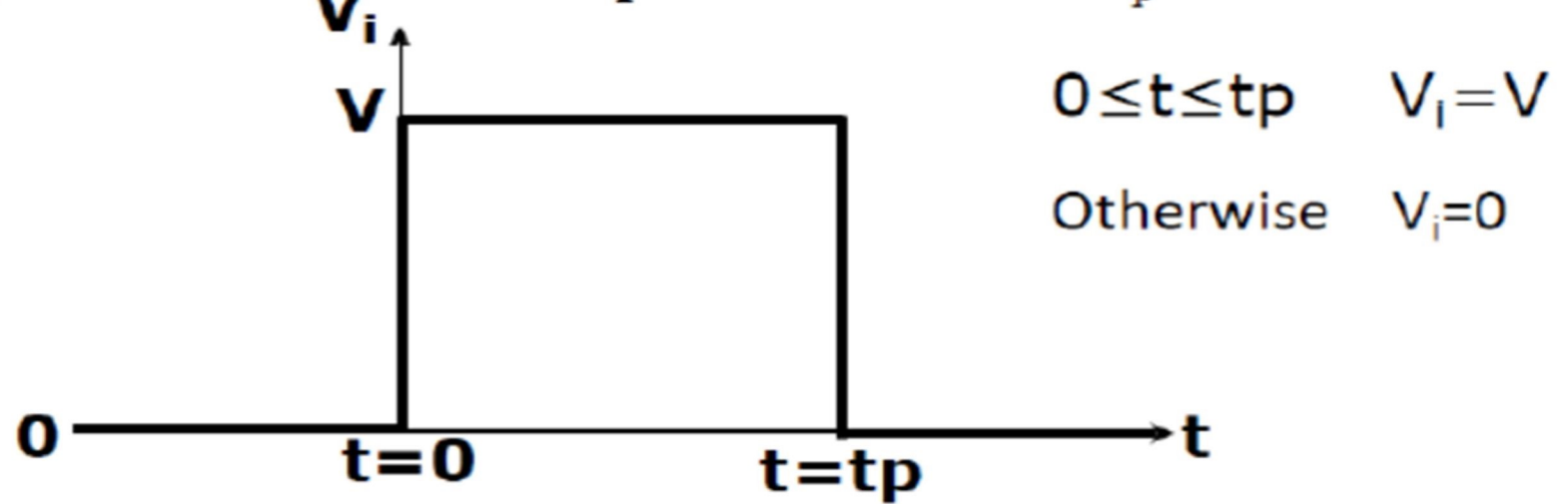
Step Waveform

A step voltage is one which maintains the value zero for all times $t < 0$ and maintains the value V for all times $t > 0$.



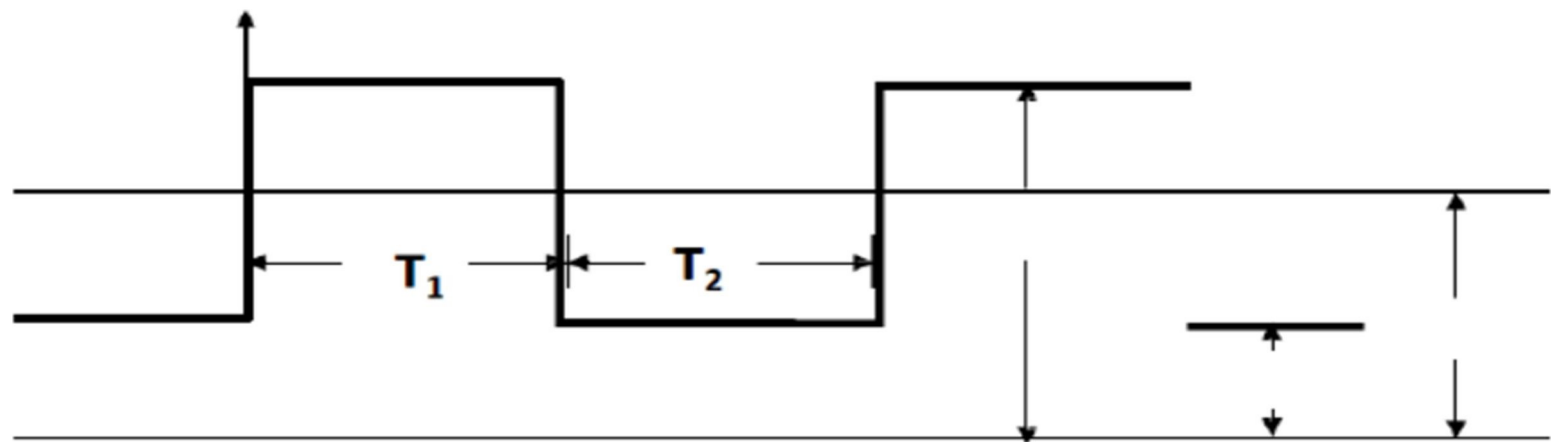
Pulse

The pulse amplitude is ' V ' and the pulse duration is t_p .



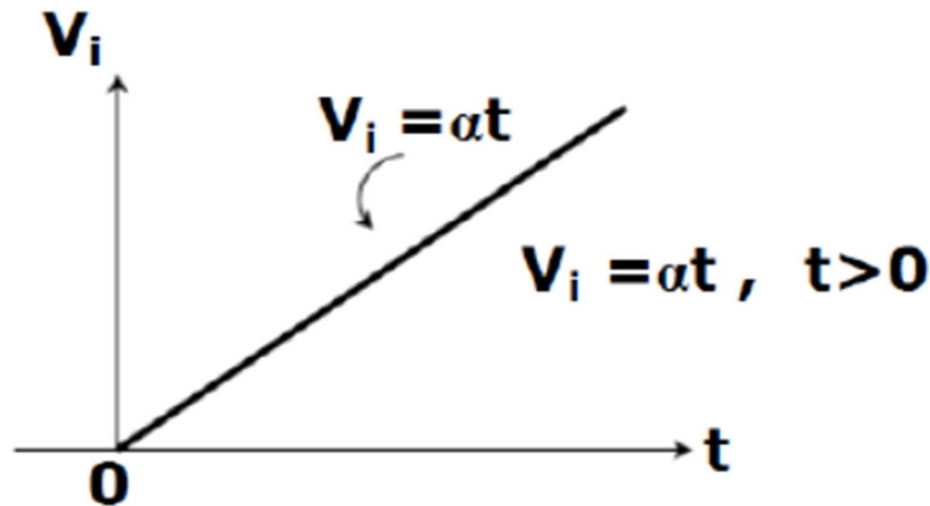
Square Wave

- A wave form which maintains itself at one constant level V^1 for a time T_1 and at other constant Level V^{11} for a time T_2 and which is repetitive with a period $T = T_1 + T_2$ is called a square-wave.



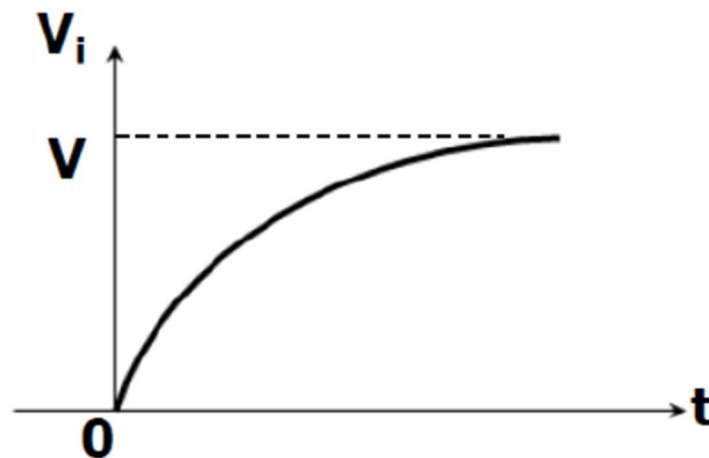
Ramp

A waveform which is zero for $t < 0$ and which increases linearly with time for $t > 0$



Exponential

- The exponential waveform input is given by $v_i = v(1 - e^{-t/T})$ where T is the time constant of the exponential input

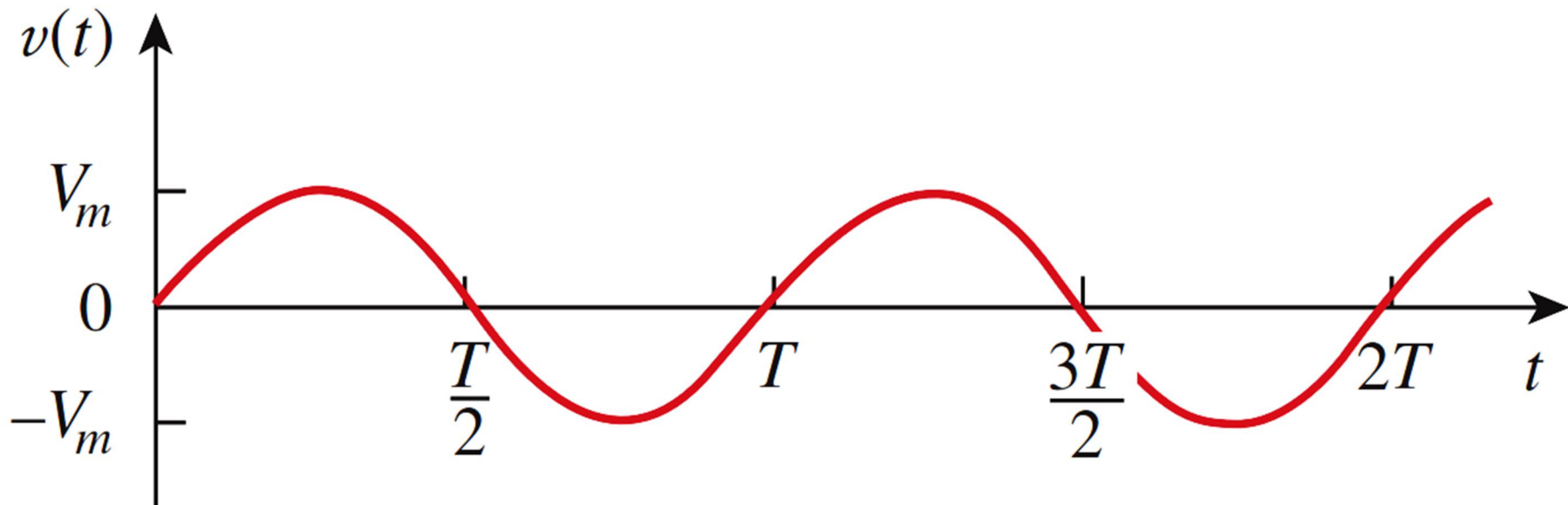
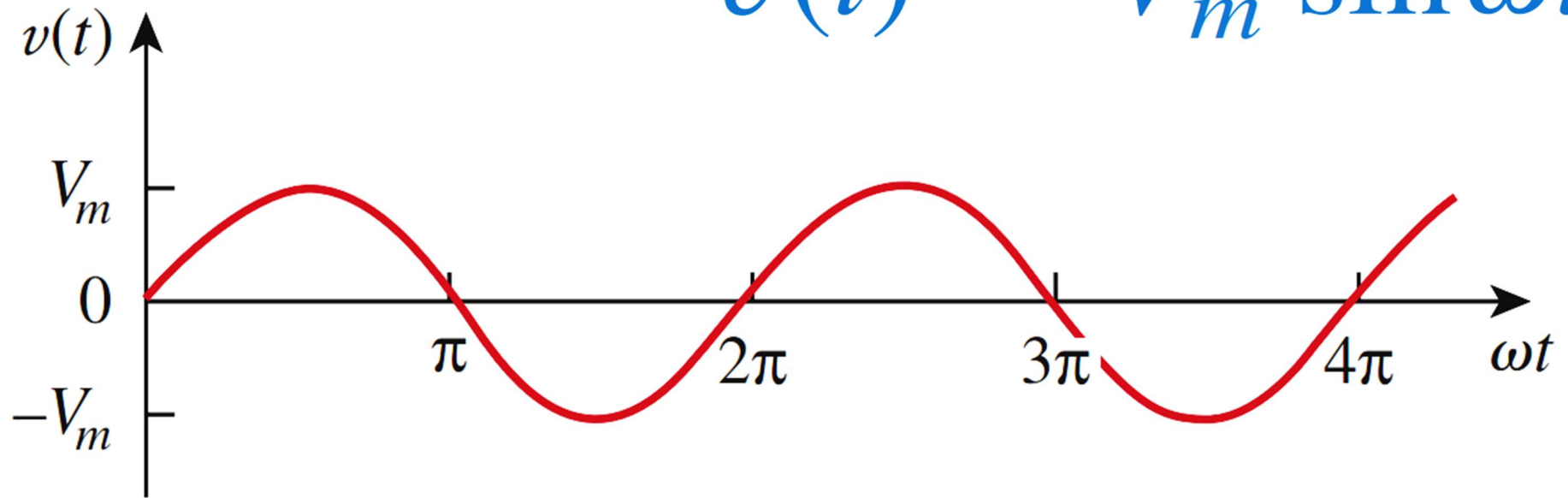


$$v_i = v(1 - e^{-t/T})$$

T = time constant of the exponential input

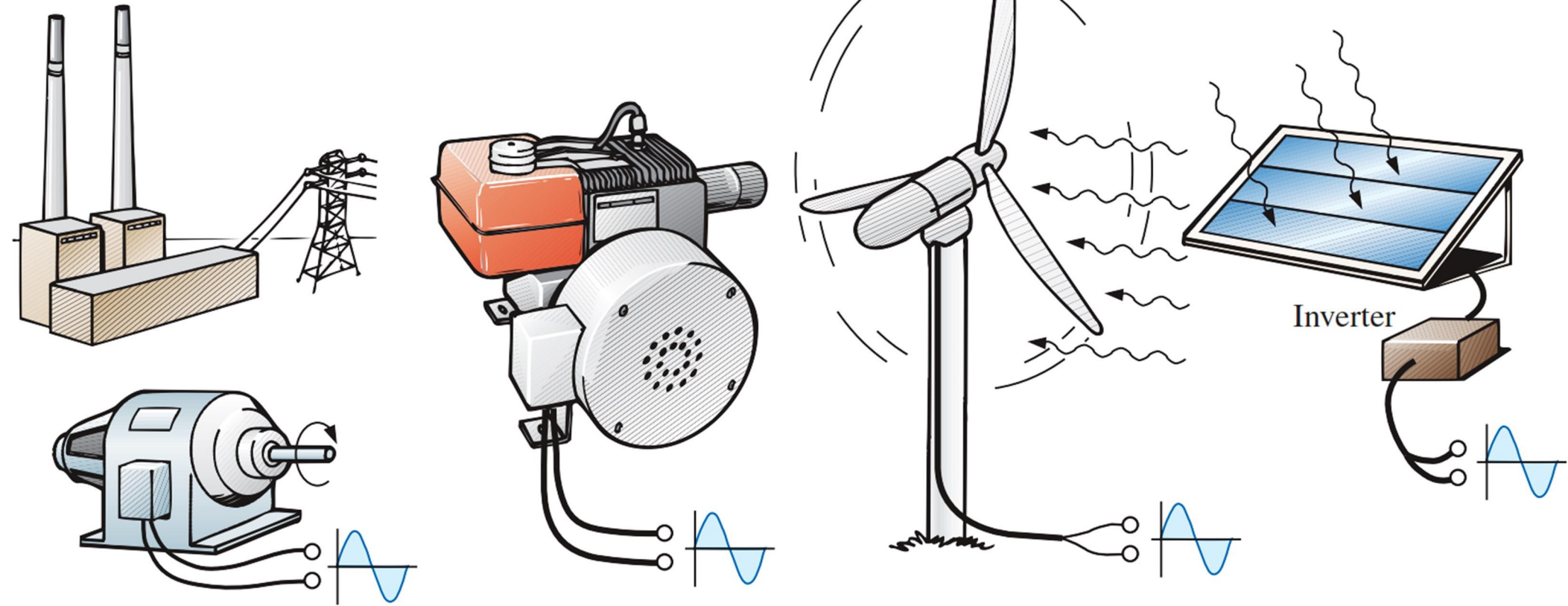
AC Voltage / Current

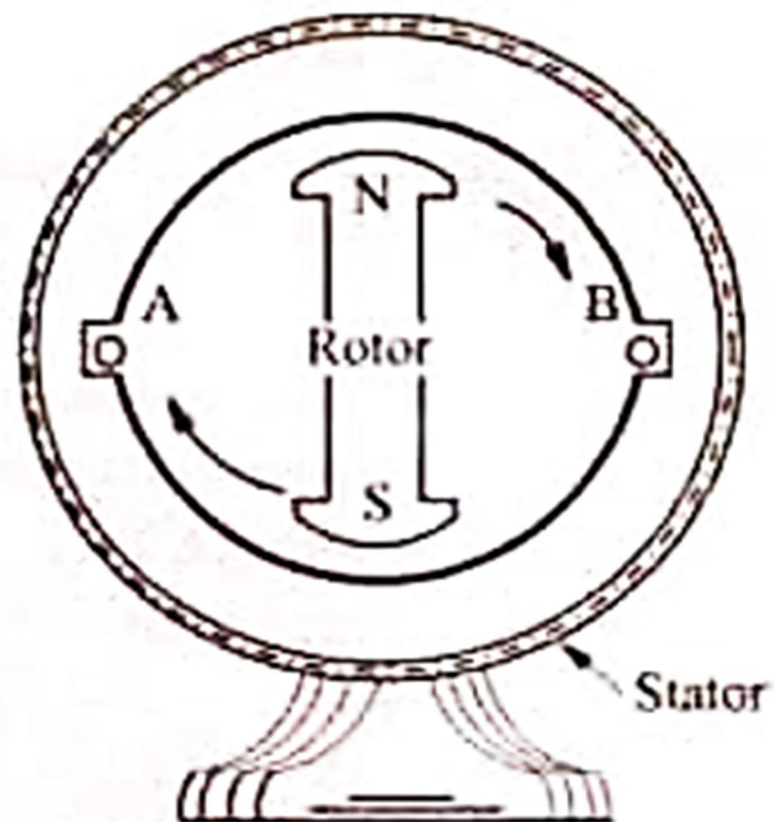
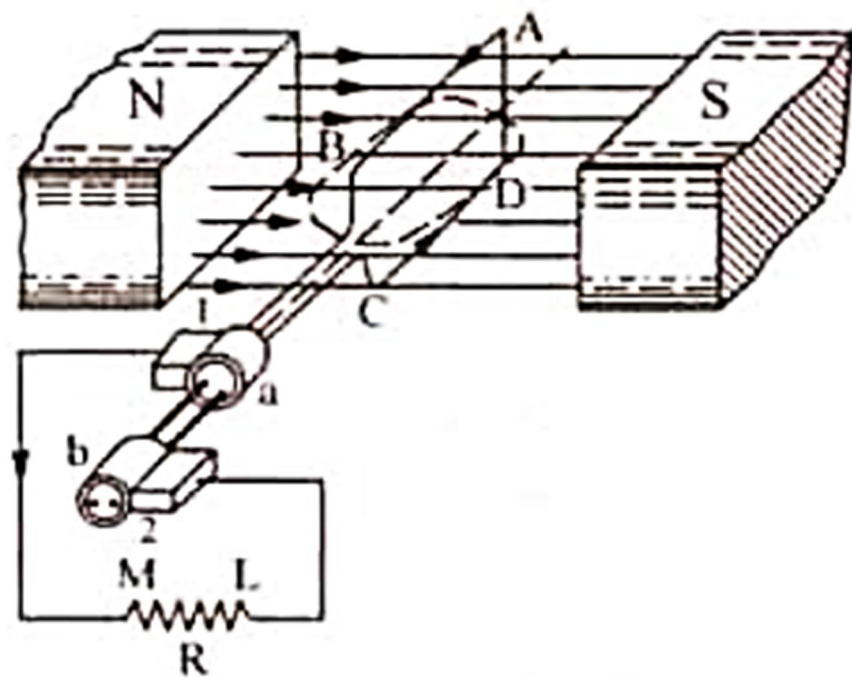
$$v(t) = V_m \sin \omega t$$



Generation of AC voltage

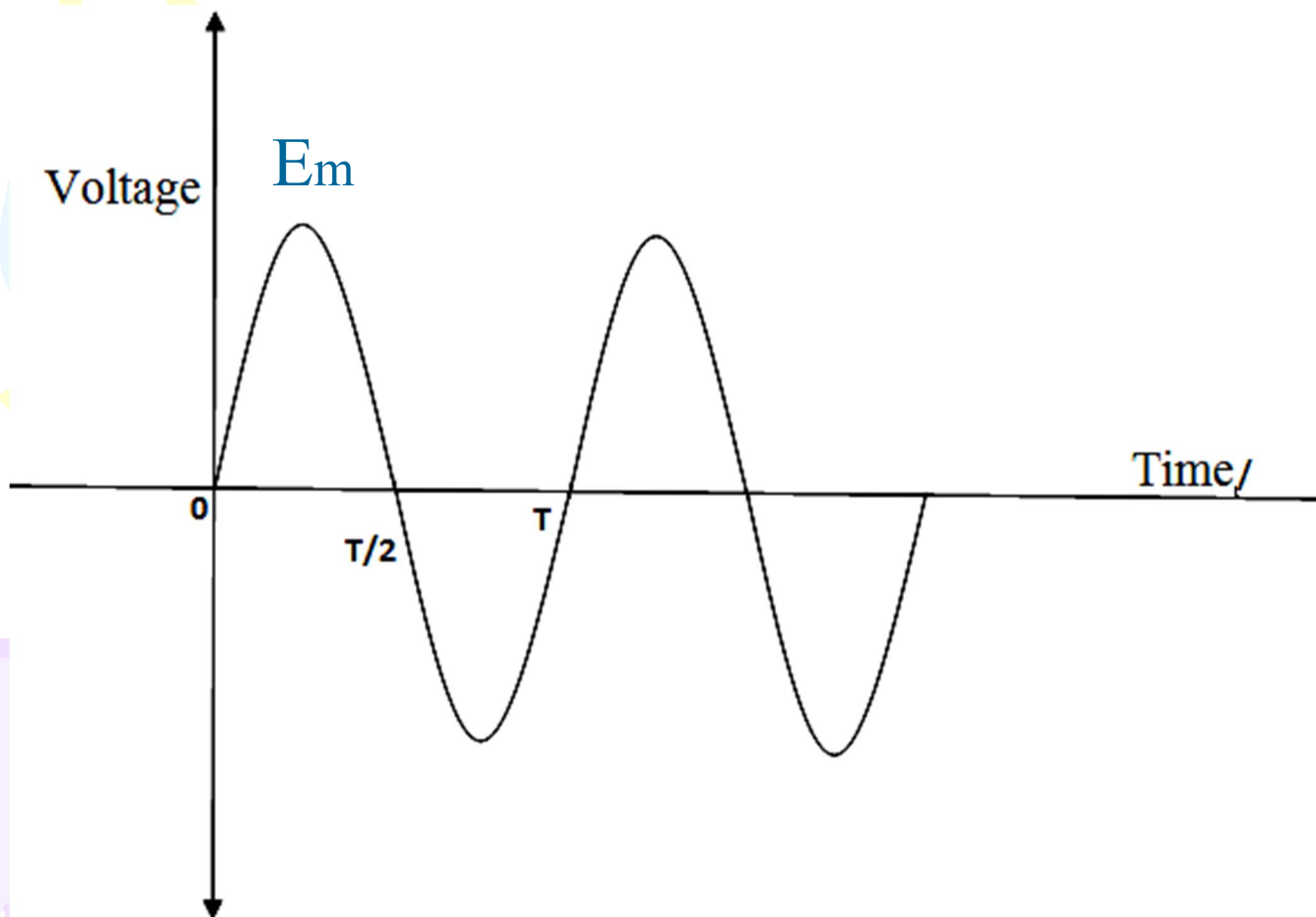
$$e = N \frac{d\phi}{dt}$$





Equation of AC voltage

$$e = E_m \sin \theta \text{ volts}$$



$$\theta = \omega t \text{ radians}$$

$$e = E_m \sin (\omega t)$$

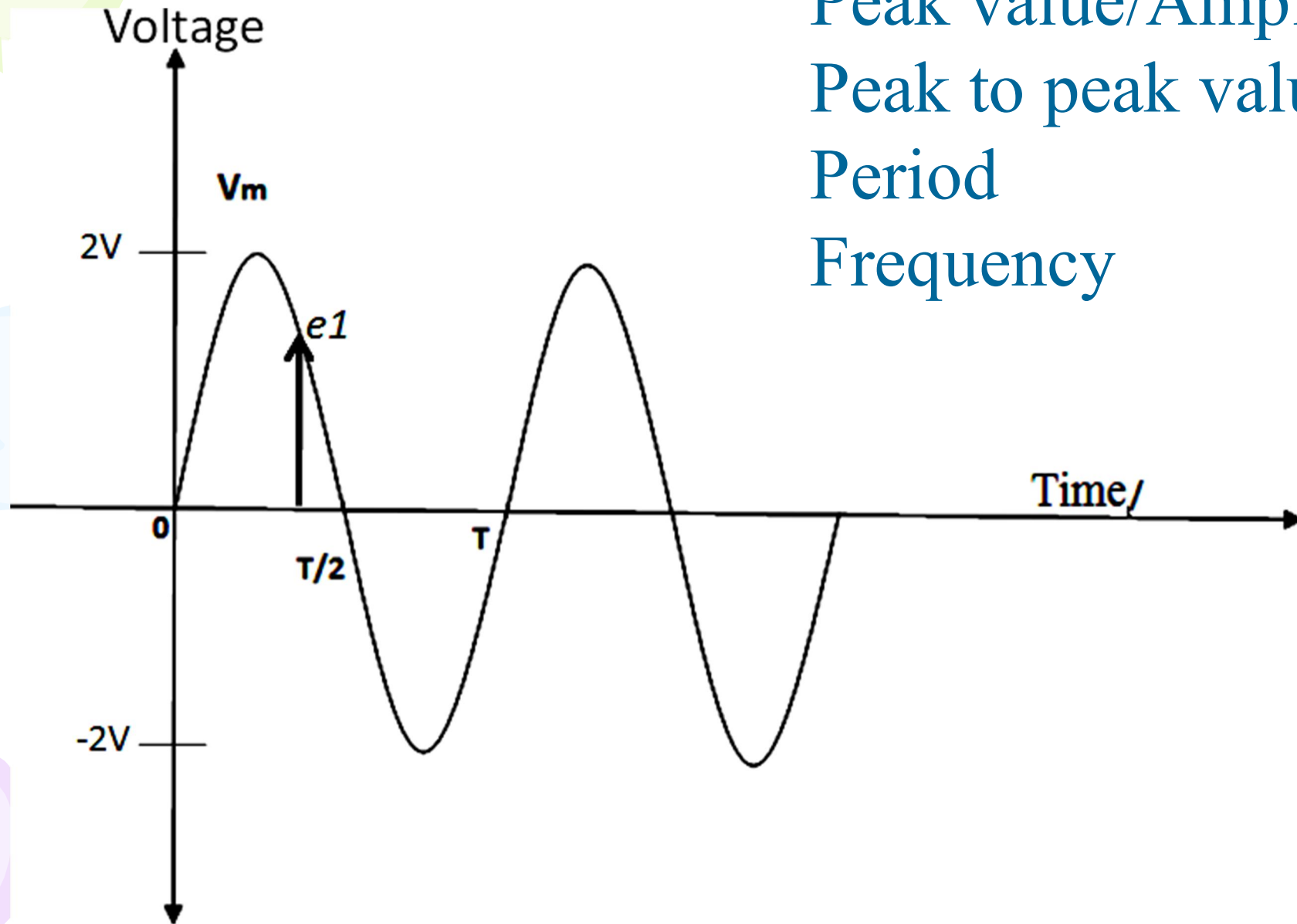
$$\omega = 2 \pi f \text{ rad / sec.}$$

$$e = E_m \sin (2 \pi f t)$$

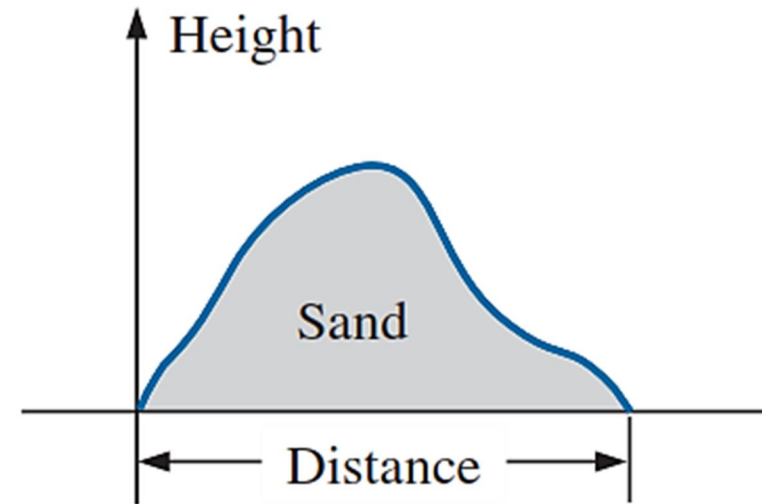
$$f = \frac{1}{T} \text{ seconds}$$

$$e = E_m \sin \left(\frac{2\pi}{T} t \right)$$

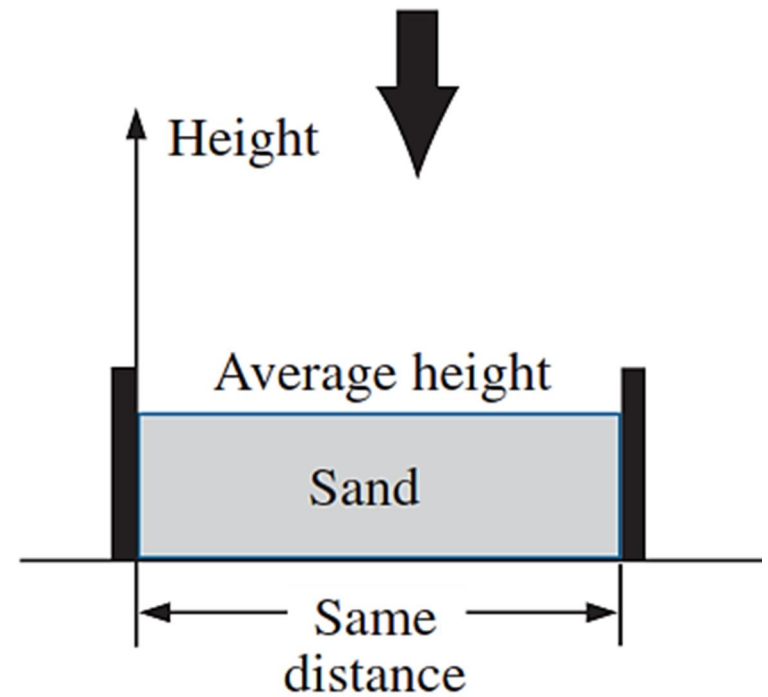
Instantaneous value
Peak value/Amplitude
Peak to peak value
Period
Frequency



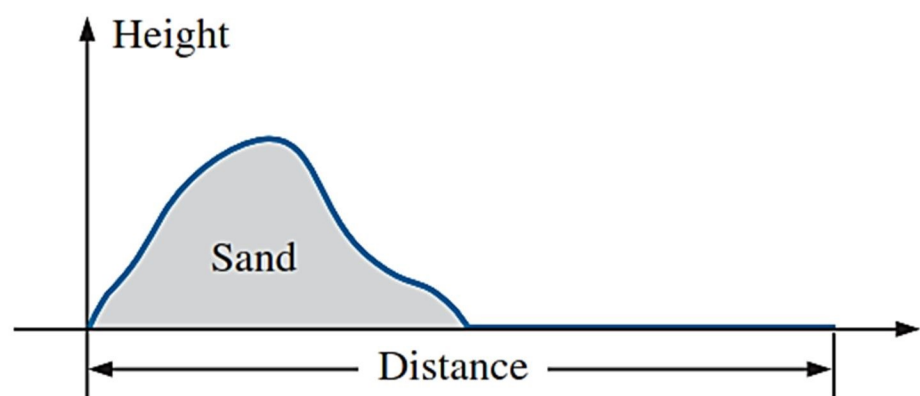
Average Value



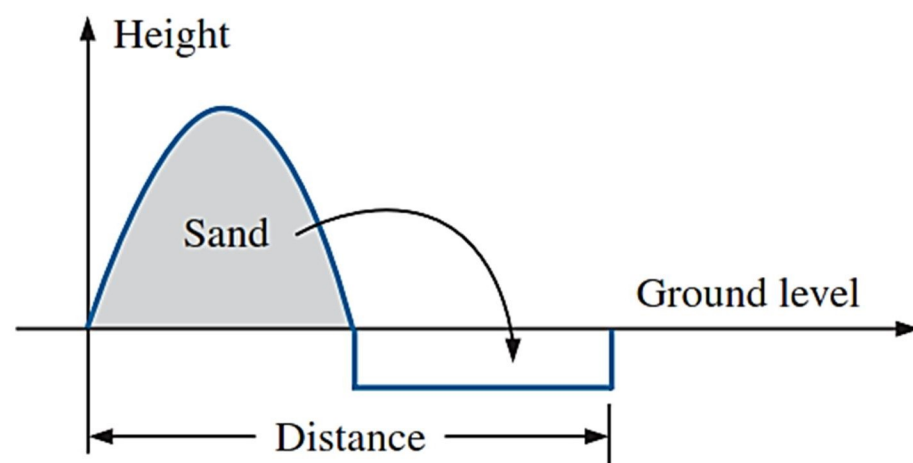
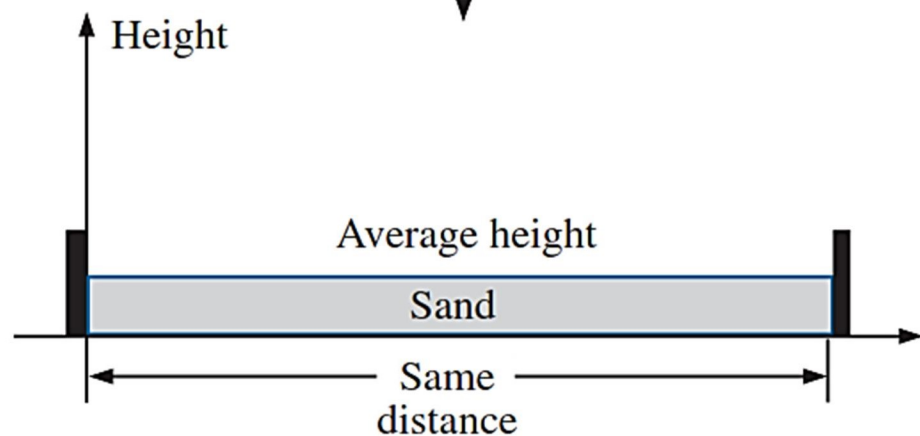
(a)



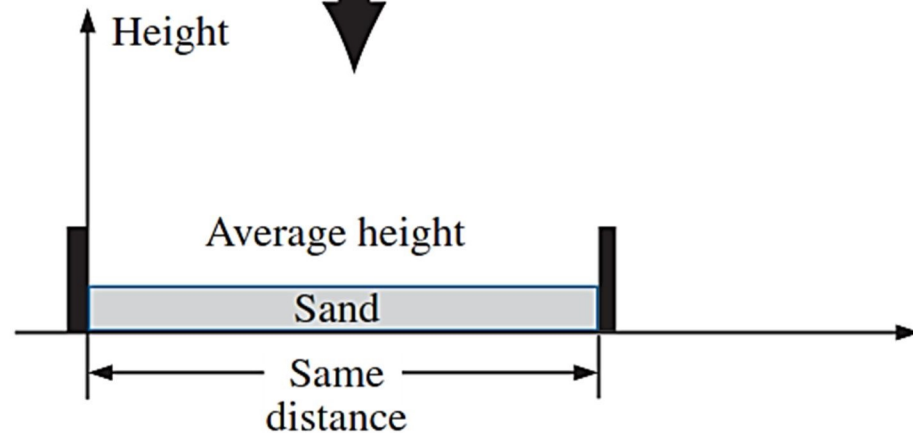
(b)



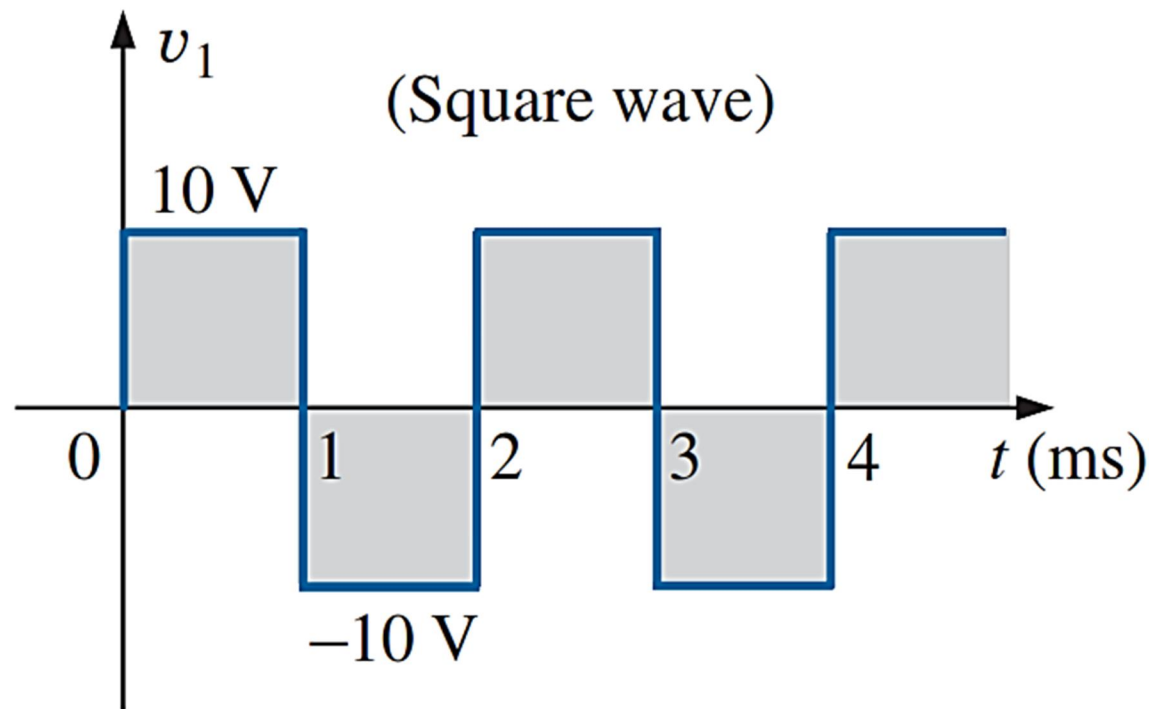
(a)



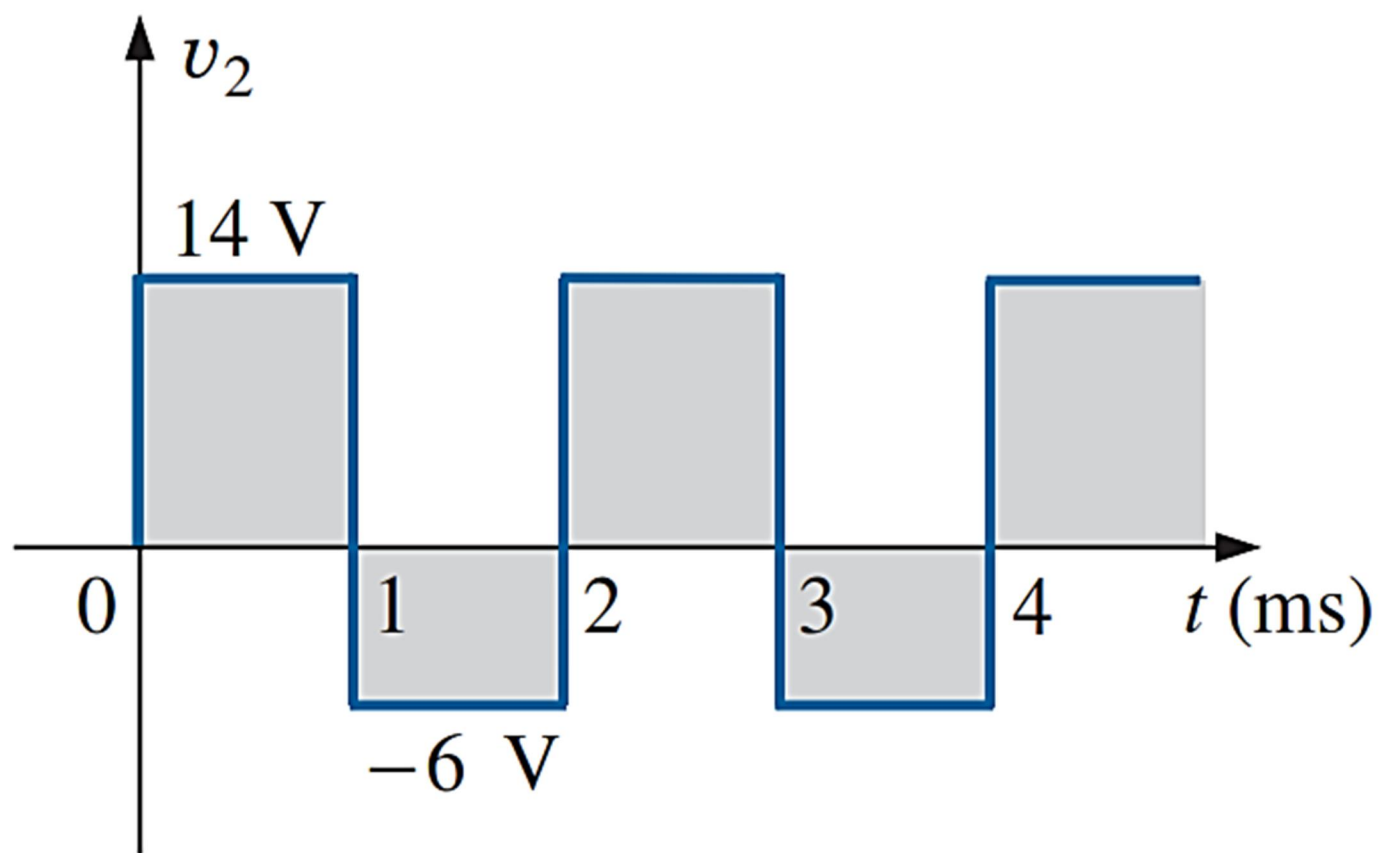
(a)



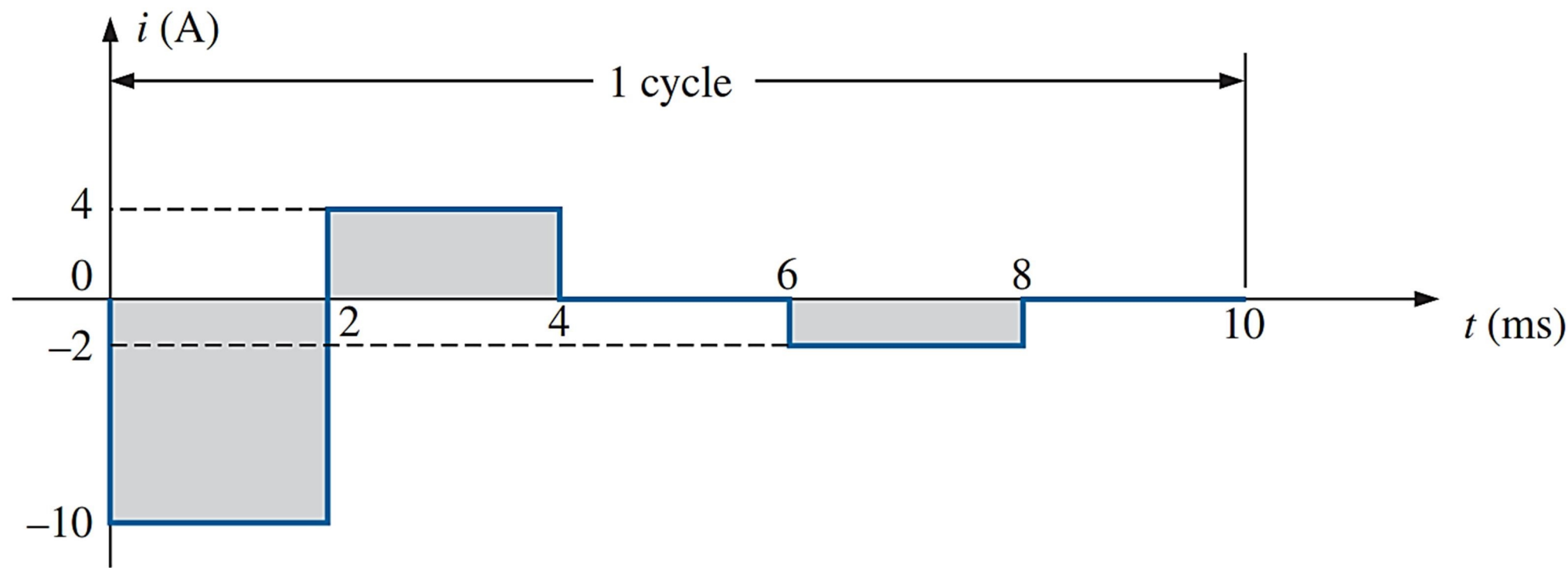
Determine the average value of the waveforms



$$G = \frac{(10 \text{ V})(1 \text{ ms}) - (10 \text{ V})(1 \text{ ms})}{2 \text{ ms}}$$
$$= \frac{0}{2 \text{ ms}} = 0 \text{ V}$$



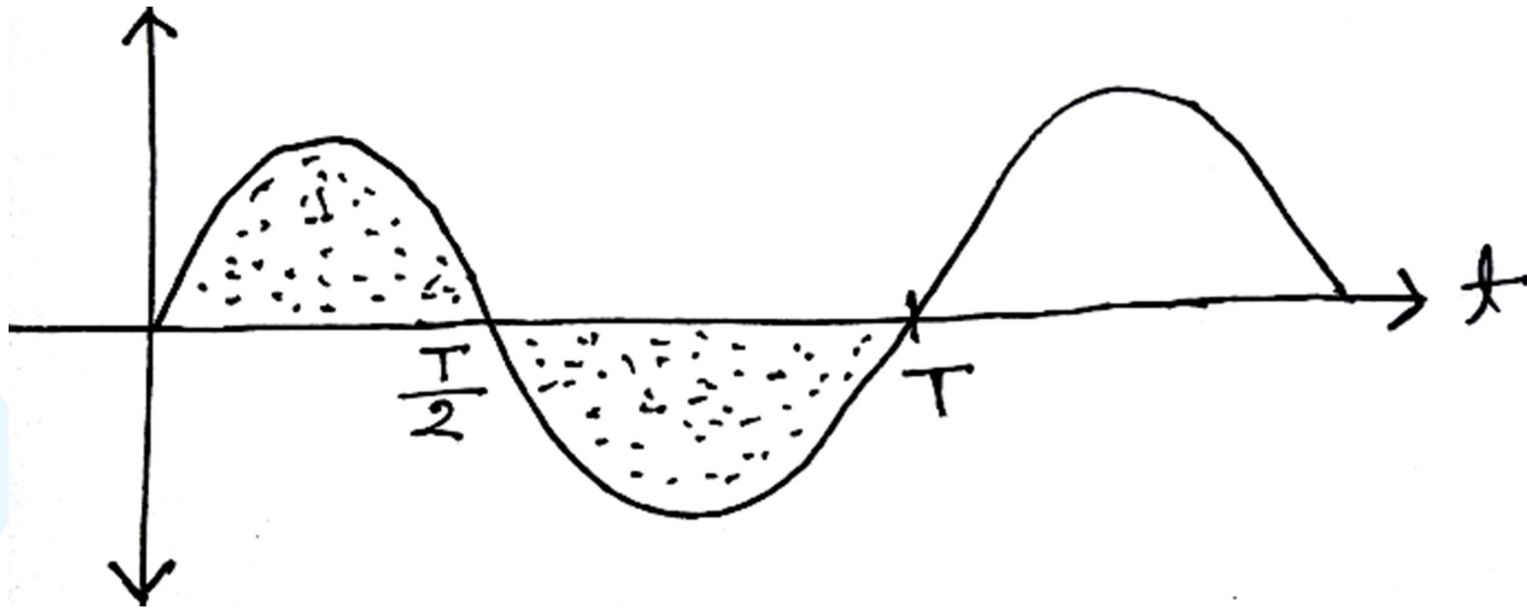
$$\begin{aligned} G &= \frac{(14 \text{ V})(1 \text{ ms}) - (6 \text{ V})(1 \text{ ms})}{2 \text{ ms}} \\ &= \frac{14 \text{ V} - 6 \text{ V}}{2} = \frac{8 \text{ V}}{2} = 4 \text{ V} \end{aligned}$$



$$= \frac{-(10 \text{ V})(2 \text{ ms}) + (4 \text{ V})(2 \text{ ms}) - (2 \text{ V})(2 \text{ ms})}{10 \text{ ms}}$$

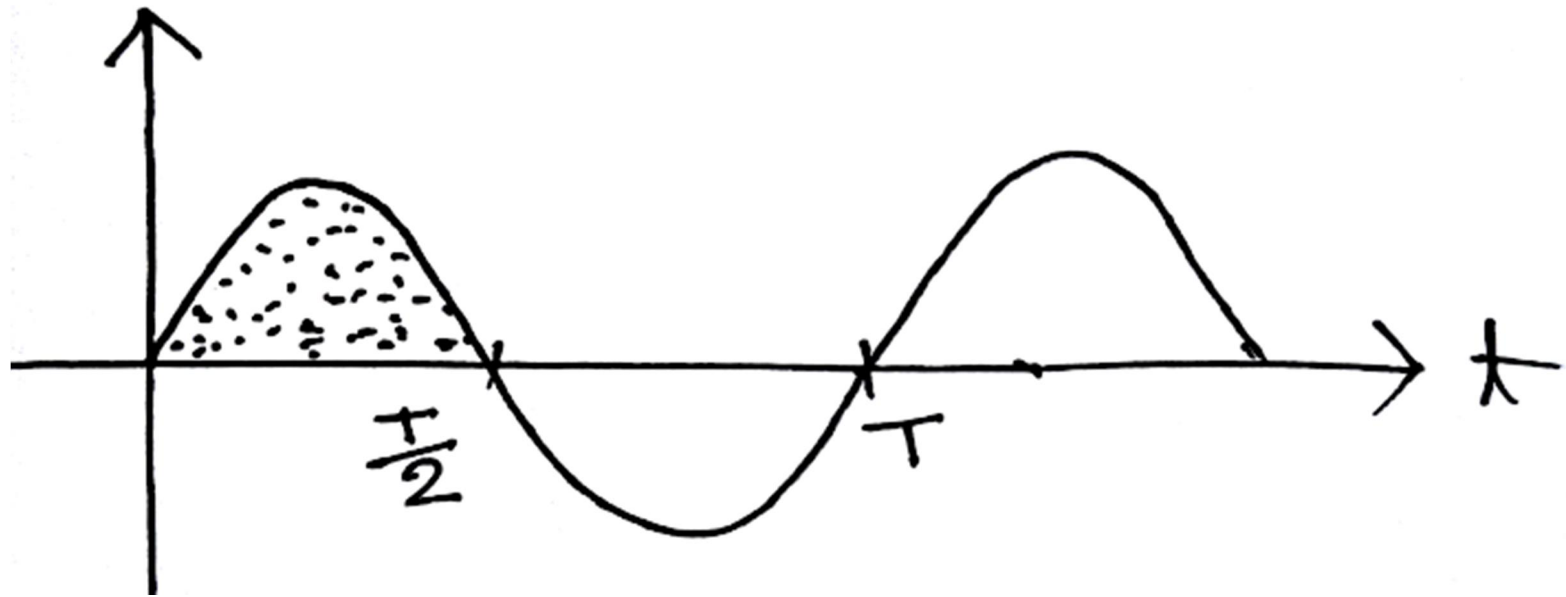
$$= \frac{-20 \text{ V} + 8 \text{ V} - 4 \text{ V}}{10} = -\frac{16 \text{ V}}{10} = -1.6 \text{ V}$$

The average value of sine wave over full-cycle



$$I_{avg} = \frac{1}{T} \int_0^T I_m \sin \omega t \, dt$$
$$= 0$$

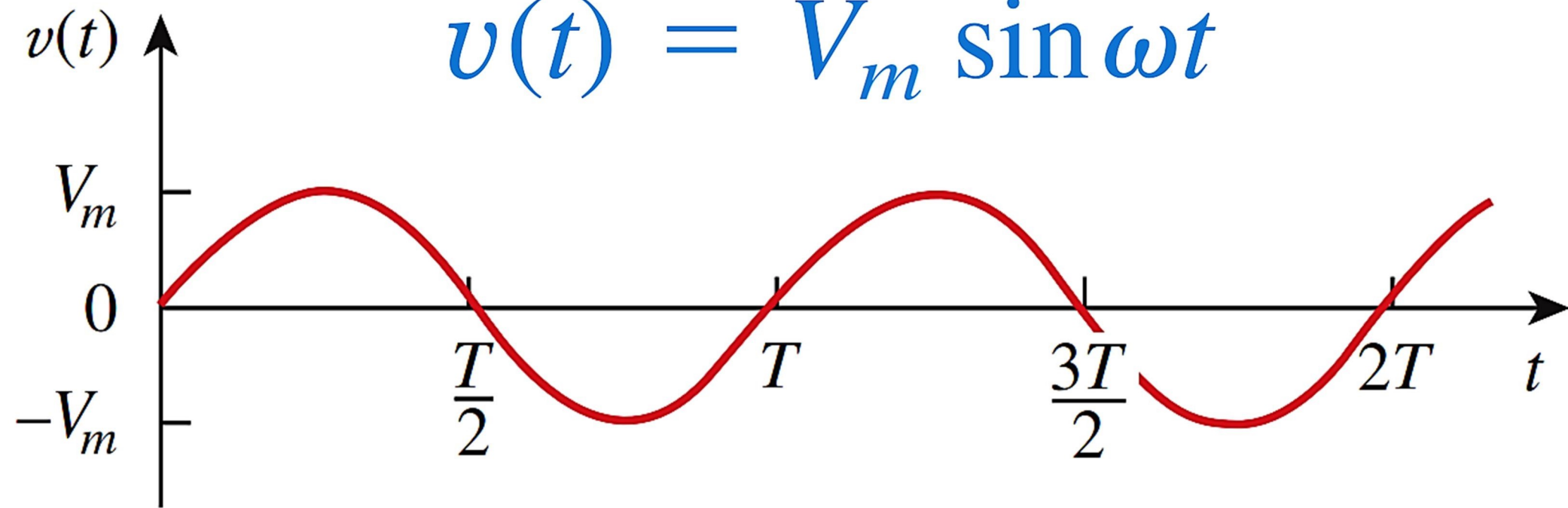
The average value of sine wave over half-cycle



$$I_{avg} = \frac{1}{\frac{T}{2}} \int_0^{T/2} I_m \sin \omega t dt$$
$$= \frac{2I_m}{\pi}$$

Effective (RMS) Value

$$v(t) = V_m \sin \omega t$$

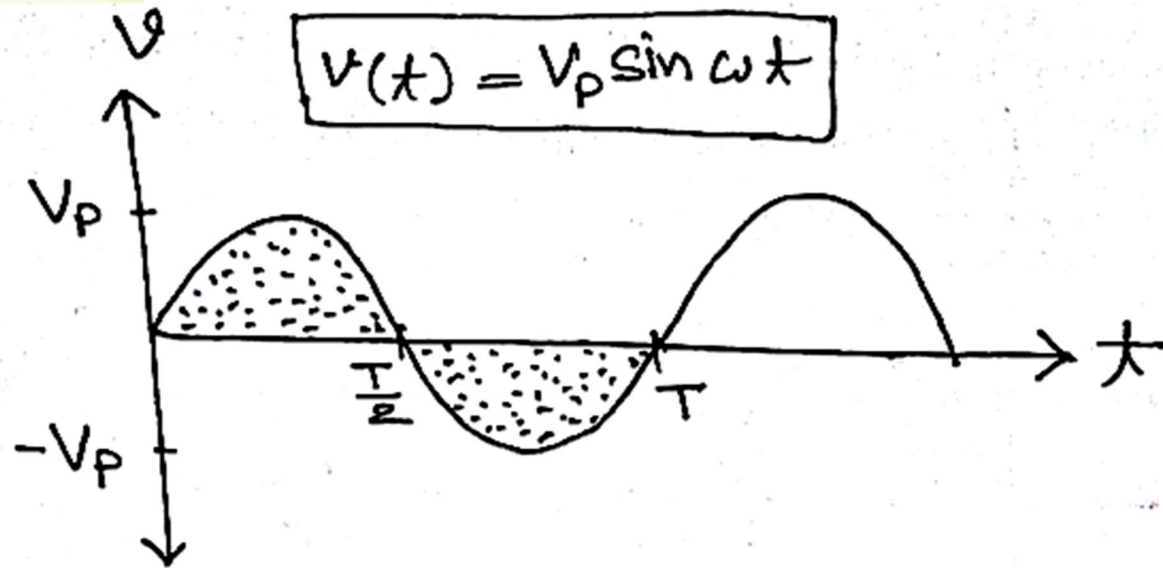


The idea of effective value arises from the need to measure the effectiveness of a voltage or current source in delivering power to a resistive load.


$$V_{\text{eff}} = \sqrt{\frac{1}{T} \int_0^T v^2 dt}$$

$$I_{\text{eff}} = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

The RMS value of sine wave



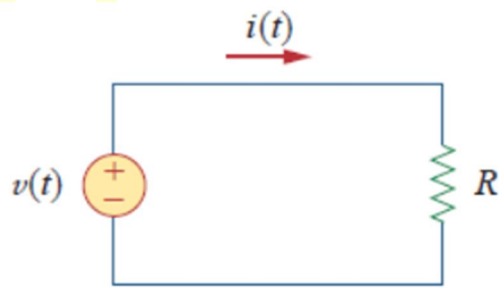
$$\begin{aligned} V_{rms} &= \sqrt{\left\{ \frac{1}{T} \int_0^T V(t) dt \right\}} \\ &= \frac{V_p}{\sqrt{2}} \\ &= \frac{\text{peak value}}{\sqrt{2}} \end{aligned}$$

DC value is equivalent to RMS value (AC)

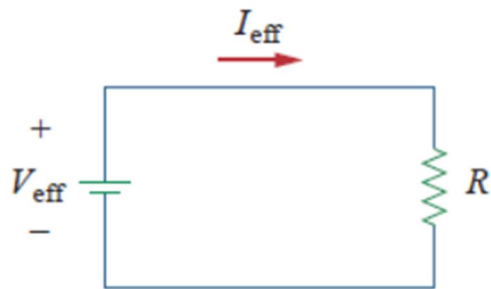
Effective or RMS Value

The idea of *effective value* arises from the need to measure the effectiveness of a voltage or current source in delivering power to a resistive load.

The **effective value** of a periodic current is the dc current that delivers the same average power to a resistor as the periodic current.



(a)



(b)

Figure 11.13: Finding the effective current: (a) ac circuit, (b) dc circuit.

Equation 11.27 states that to find the rms value of $x(t)$, we first find its *square* x^2 and then find the *mean* of that, or

$$\frac{1}{T} \int_0^T x^2 dt$$

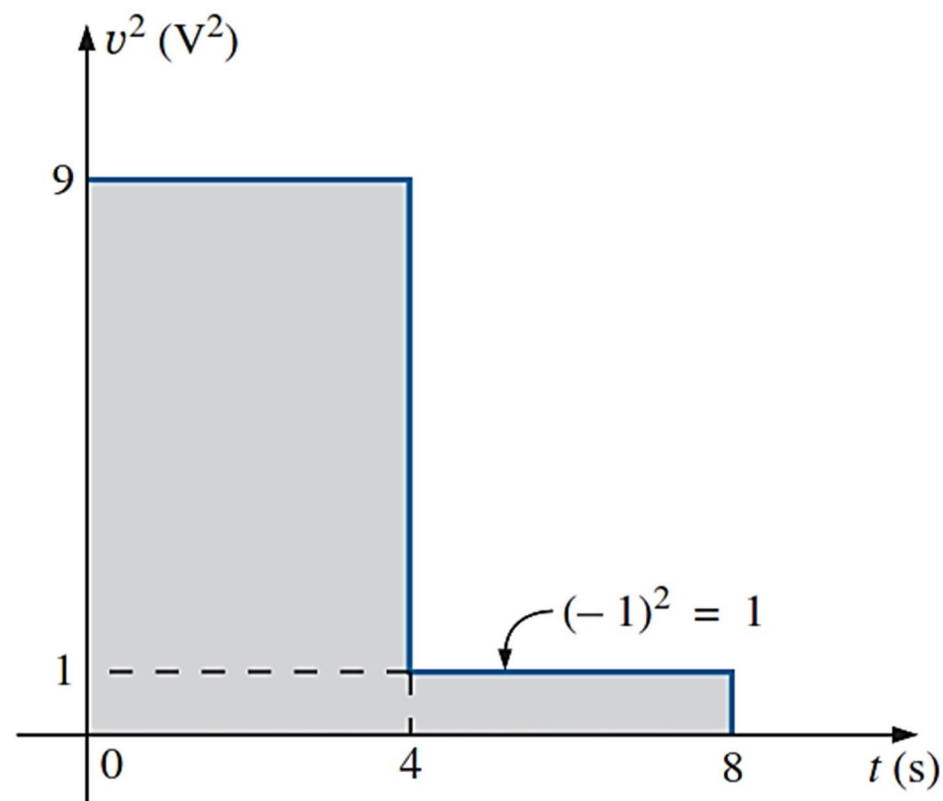
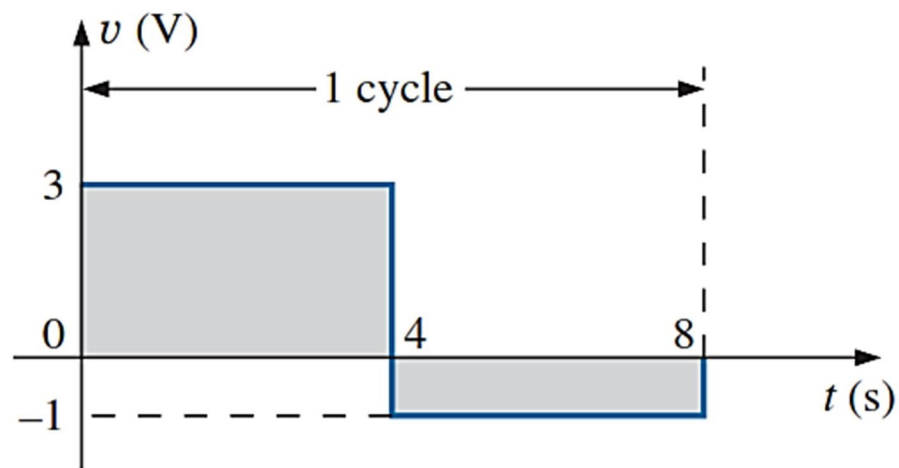
and then the square *root* ($\sqrt{\quad}$) of that mean. The rms value of a constant is the constant itself. For the sinusoid $i(t) = I_m \cos \omega t$, the effective or rms value is

$$\begin{aligned} I_{\text{rms}} &= \sqrt{\frac{1}{T} \int_0^T I_m^2 \cos^2 \omega t dt} \\ &= \sqrt{\frac{I_m^2}{T} \int_0^T \frac{1}{2} (1 + \cos 2\omega t) dt} = \frac{I_m}{\sqrt{2}} \end{aligned} \quad (11.28)$$

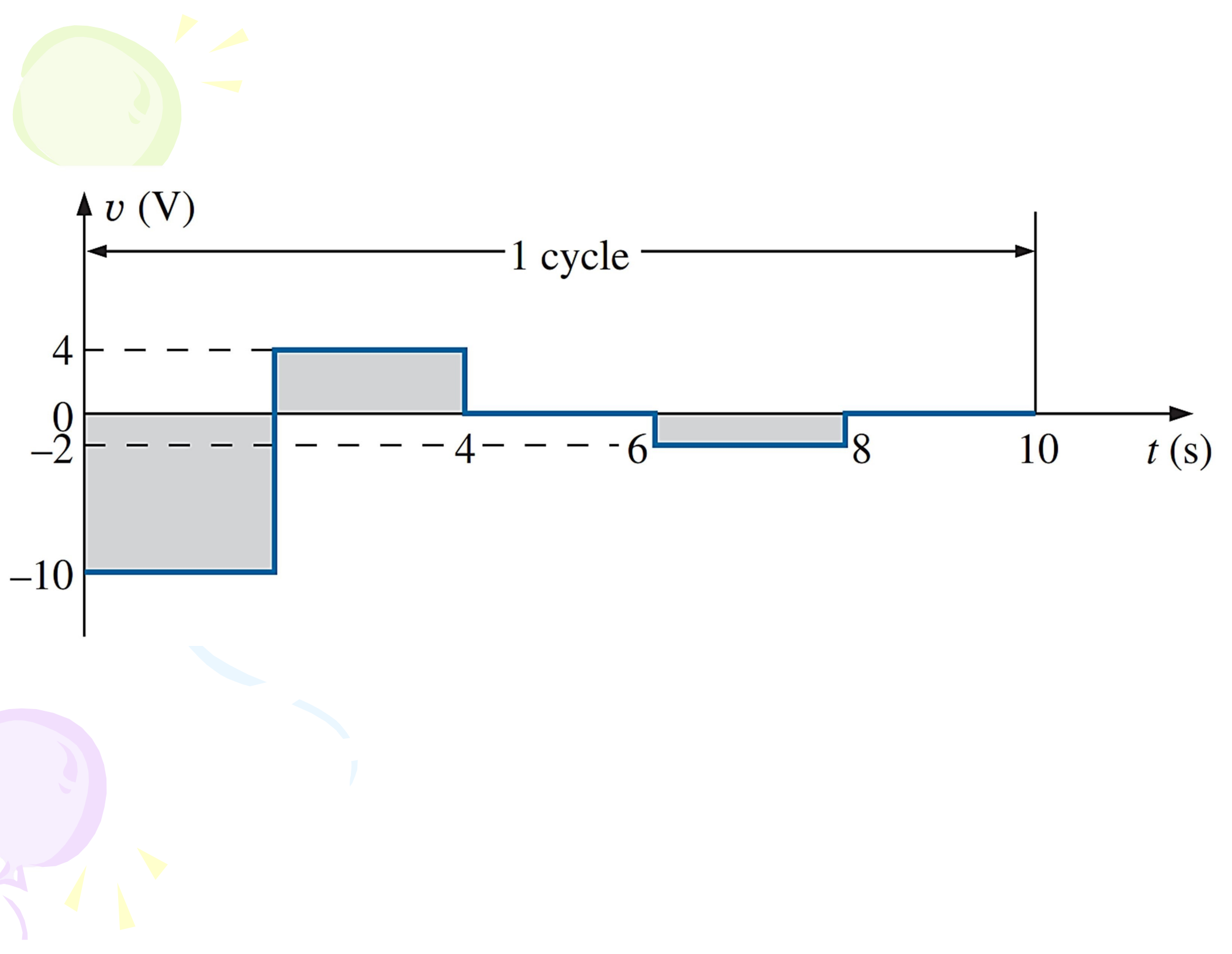
Similarly, for $v(t) = V_m \cos \omega t$,

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} \quad (11.29)$$

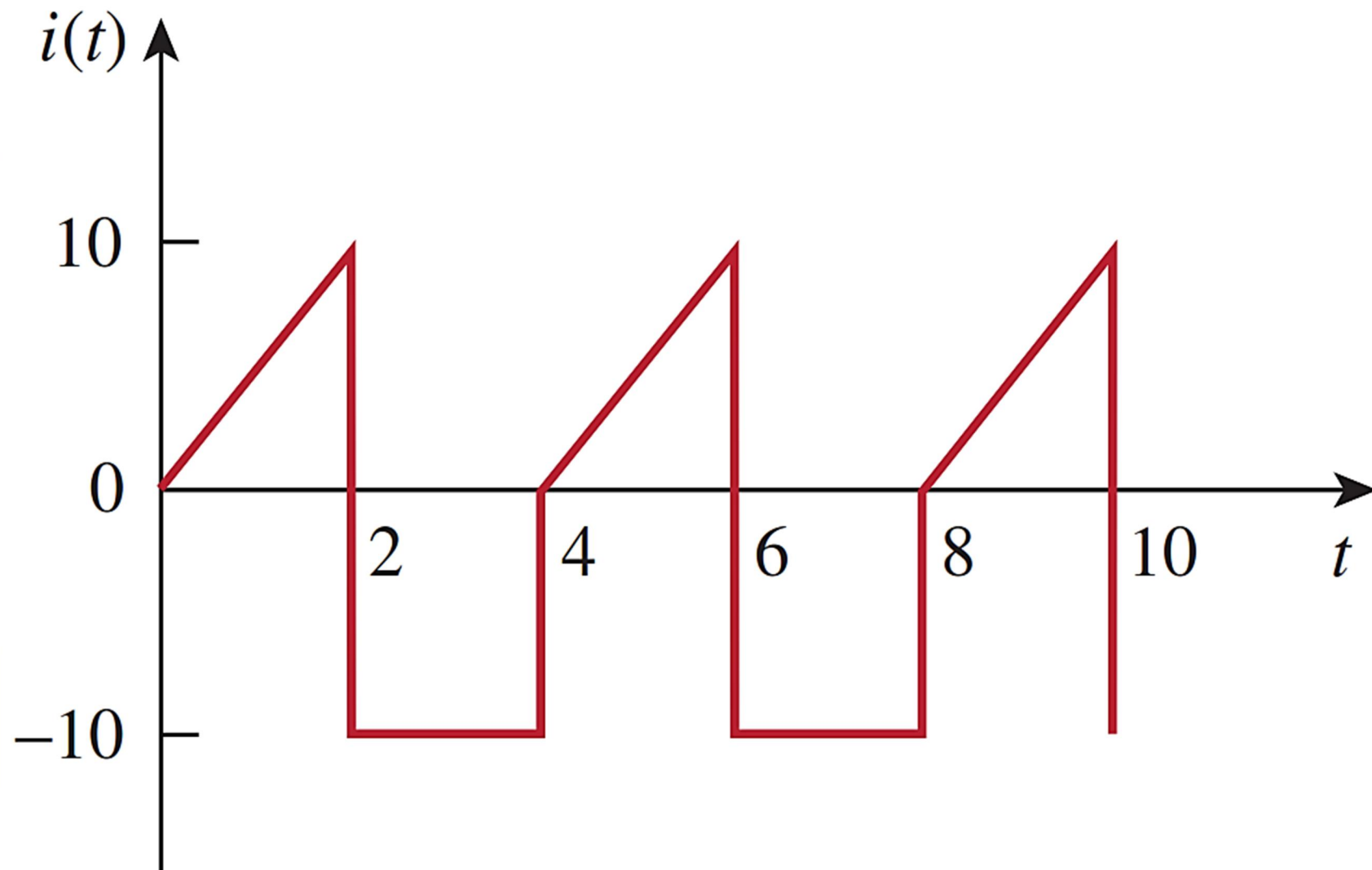
Keep in mind that Eqs. (11.28) and (11.29) are only valid for sinusoidal signals.



$$V_{\text{rms}} = \sqrt{\frac{(9)(4) + (1)(4)}{8}} = \sqrt{\frac{40}{8}} = 2.24 \text{ V}$$



Determine the rms value of the current waveform in Figure. If the current is passed through a resistor, find the average power absorbed by the resistor.




$$i(t) = \begin{cases} 5t, & 0 < t < 2 \\ -10, & 2 < t < 4 \end{cases}$$

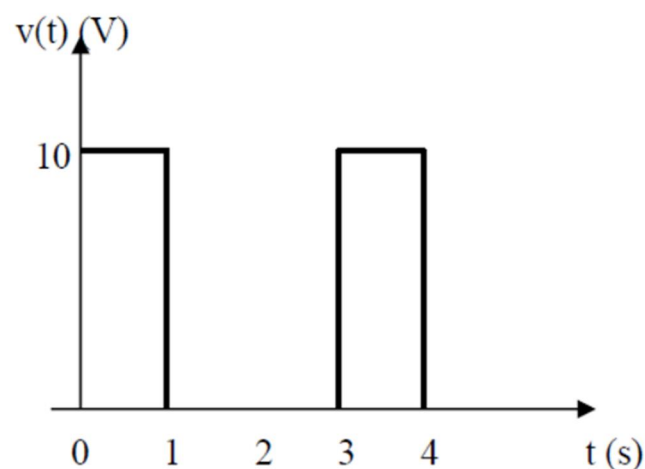
The rms value is


$$\begin{aligned} I_{\text{rms}} &= \sqrt{\frac{1}{T} \int_0^T i^2 dt} = \sqrt{\frac{1}{4} \left[\int_0^2 (5t)^2 dt + \int_2^4 (-10)^2 dt \right]} \\ &= \sqrt{\frac{1}{4} \left[25 \frac{t^3}{3} \Big|_0^2 + 100t \Big|_2^4 \right]} = \sqrt{\frac{1}{4} \left(\frac{200}{3} + 200 \right)} = 8.165 \text{ A} \end{aligned}$$

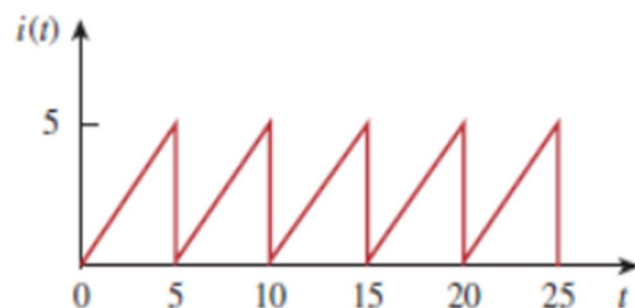
The power absorbed by a 2-Ω resistor is


$$P = I_{\text{rms}}^2 R = (8.165)^2 (2) = 133.3 \text{ W}$$

Determine the rms value of the voltage shown in Fig. 11.54.



11.27 Calculate the rms value of the current waveform of Fig. 11.58.

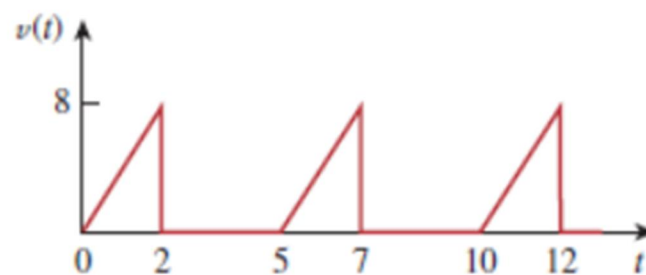


11.24 Determine the rms value of the waveform in Fig. 11.55.



Figure 11.55
For Prob. 11.24.

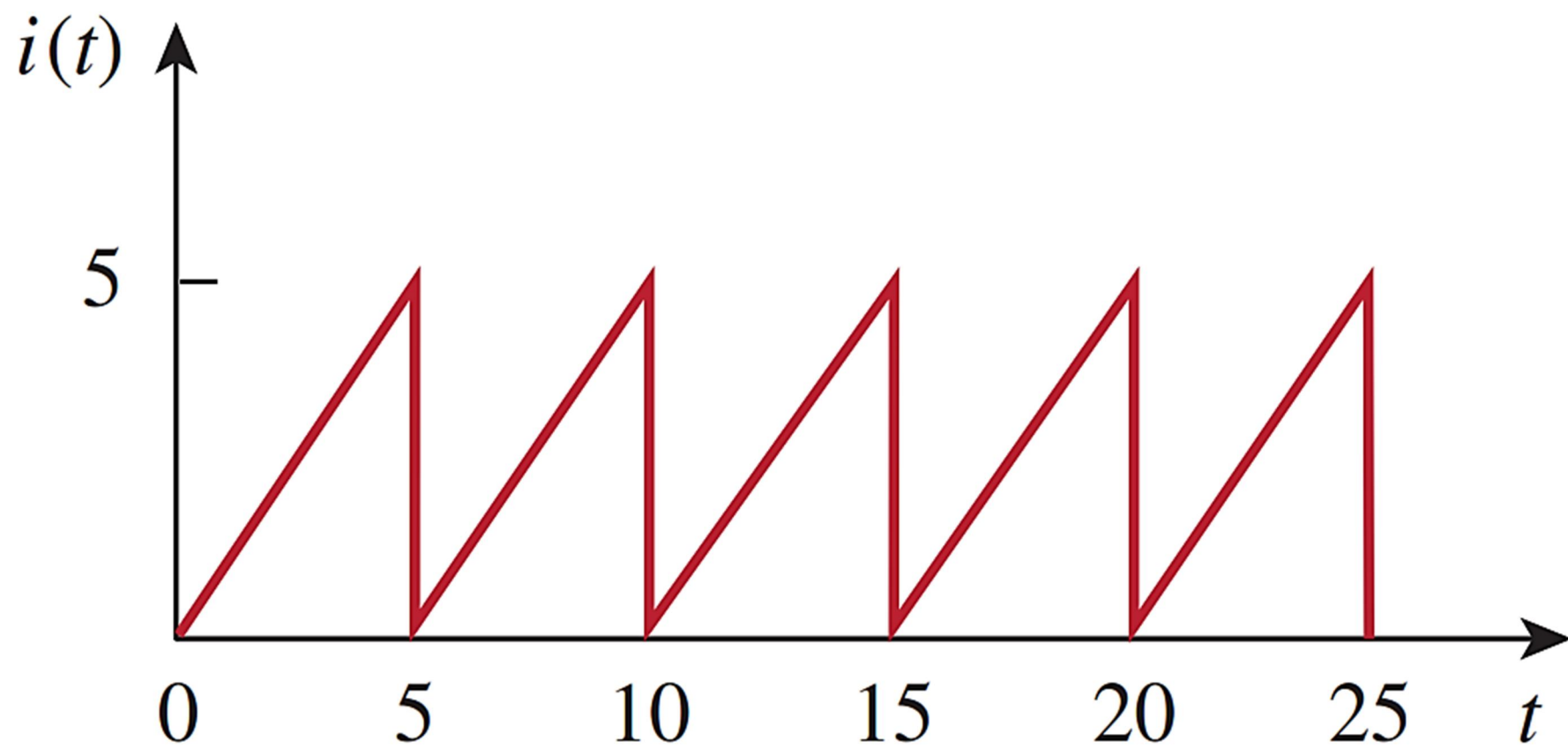
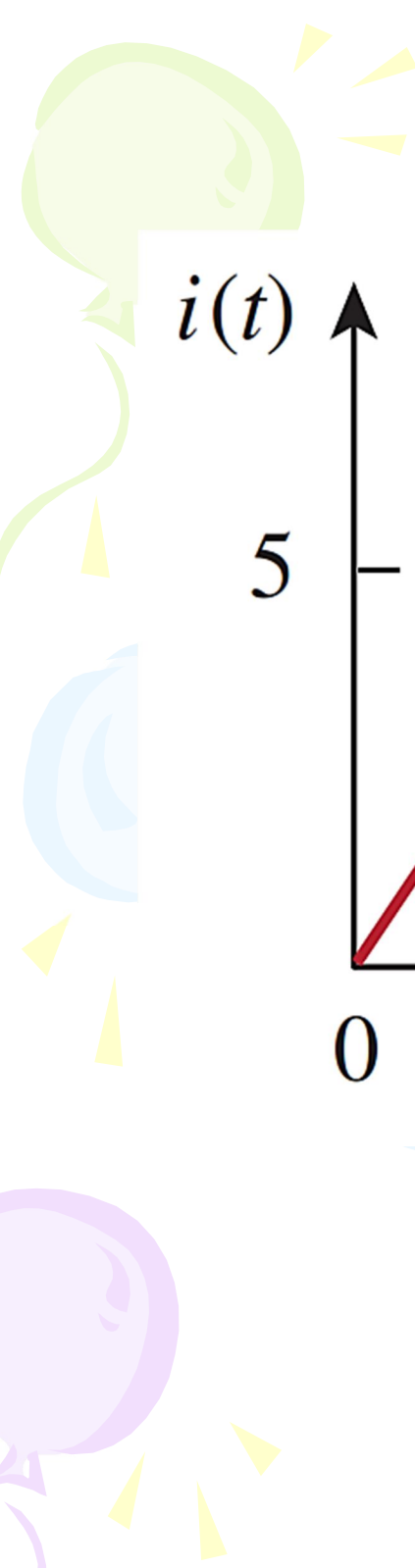
11.28 Find the rms value of the voltage waveform of Fig. 11.59 as well as the average power absorbed by a $2\text{-}\Omega$ resistor when the voltage is applied across the resistor.

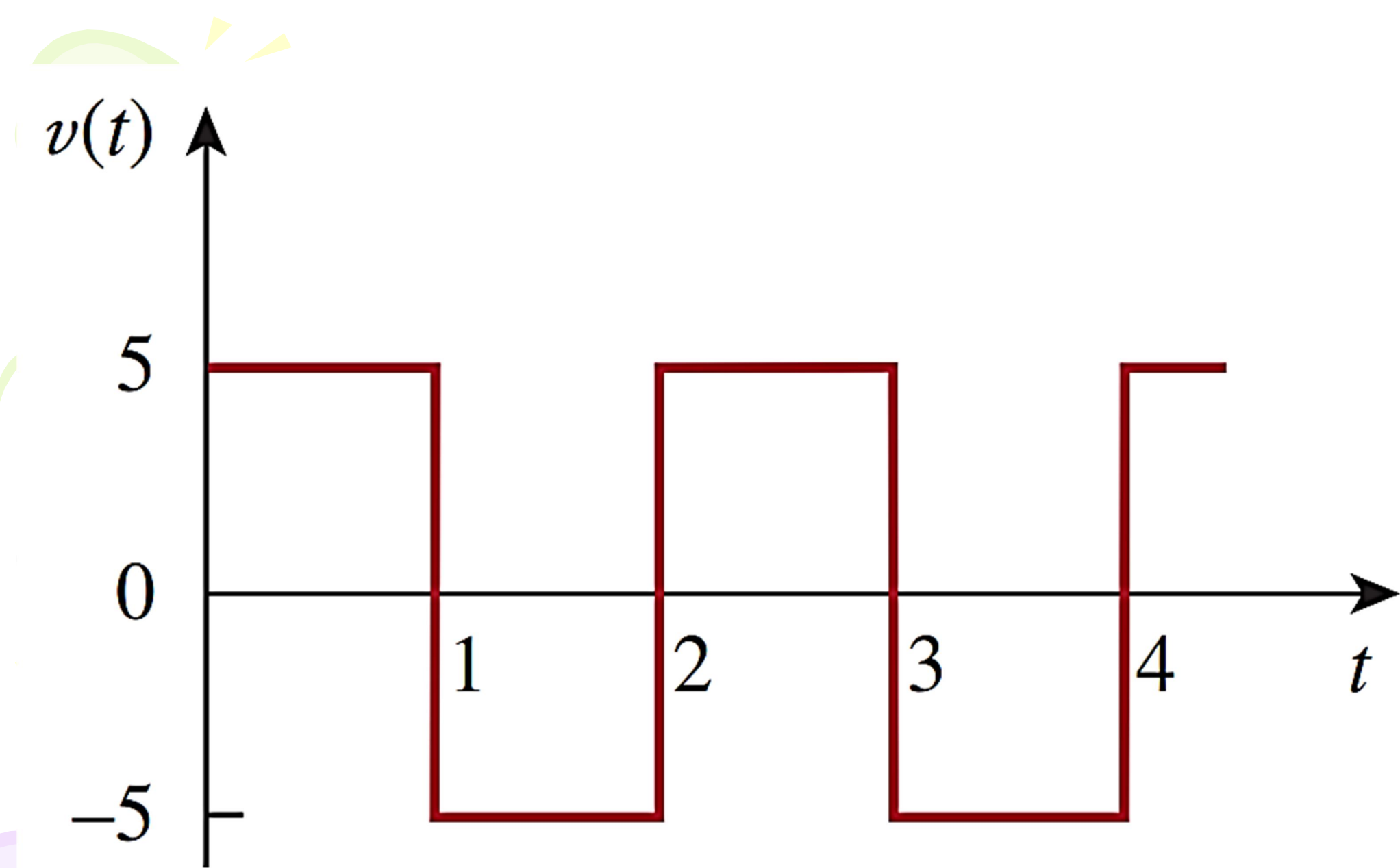


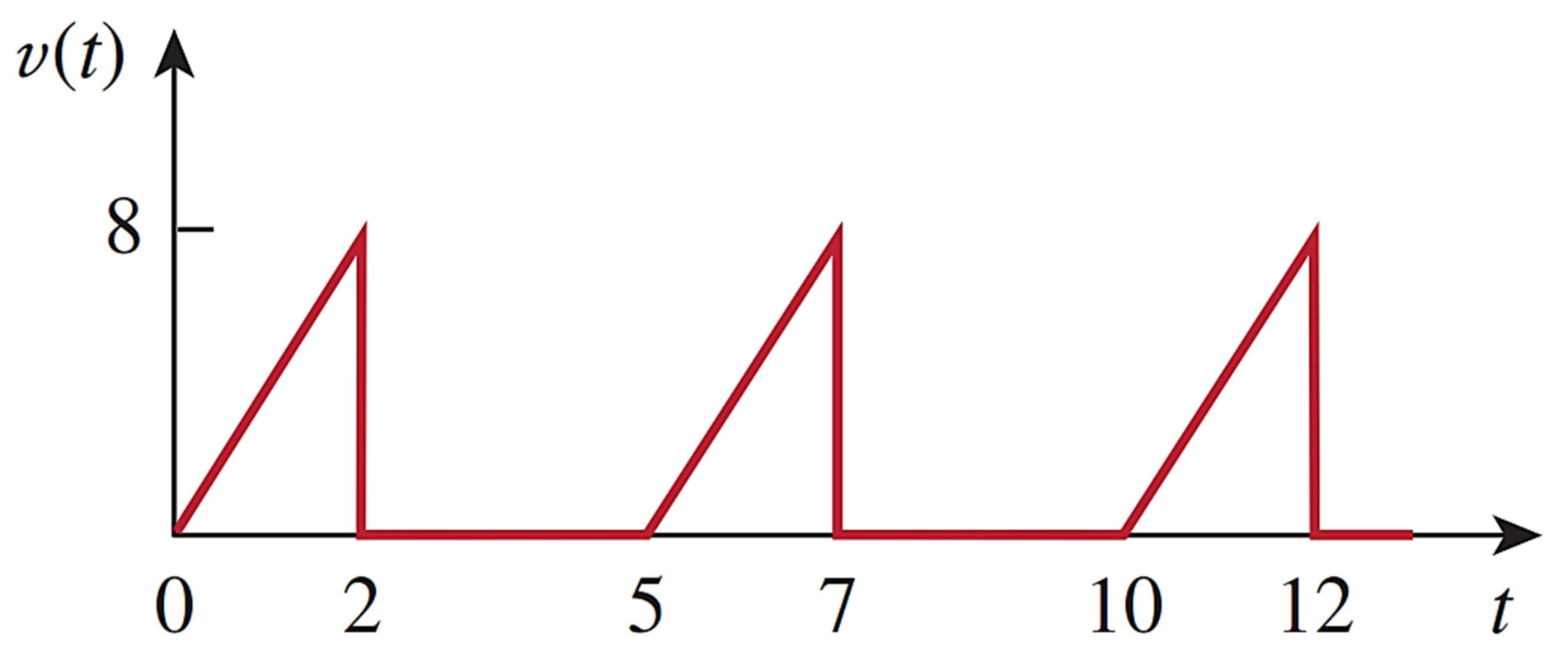
$v(t)$

V_p





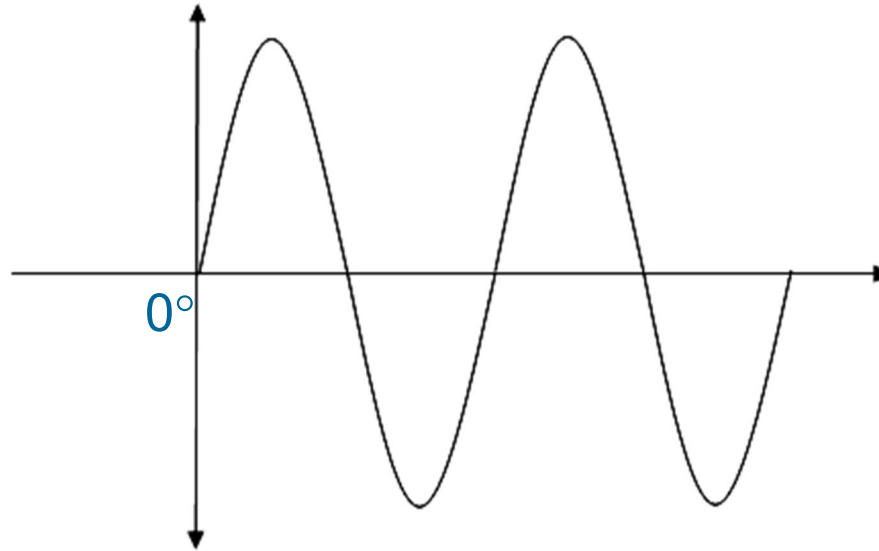




Phase

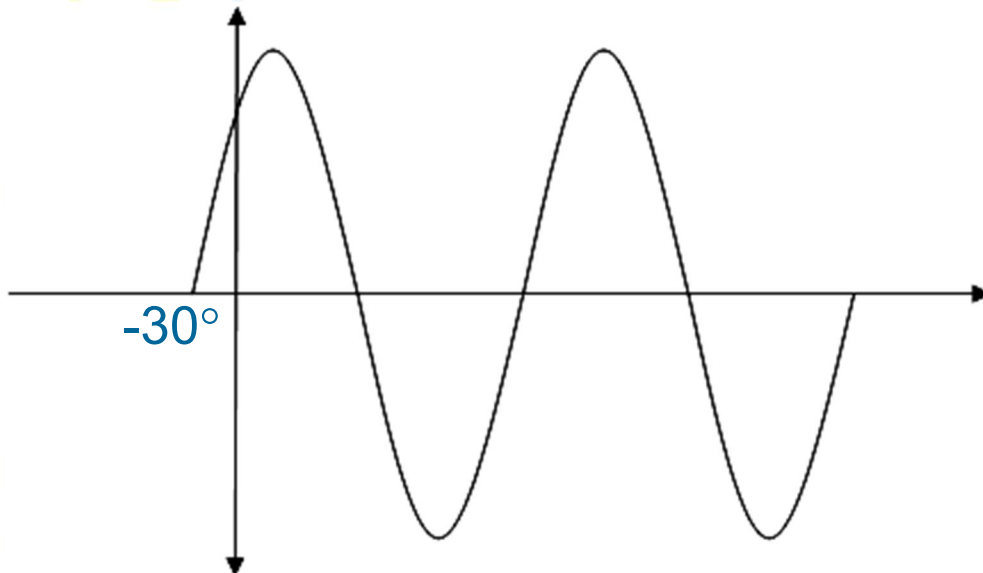
The phase is zero.

Reference
 $i = I_m \sin(\omega t)$



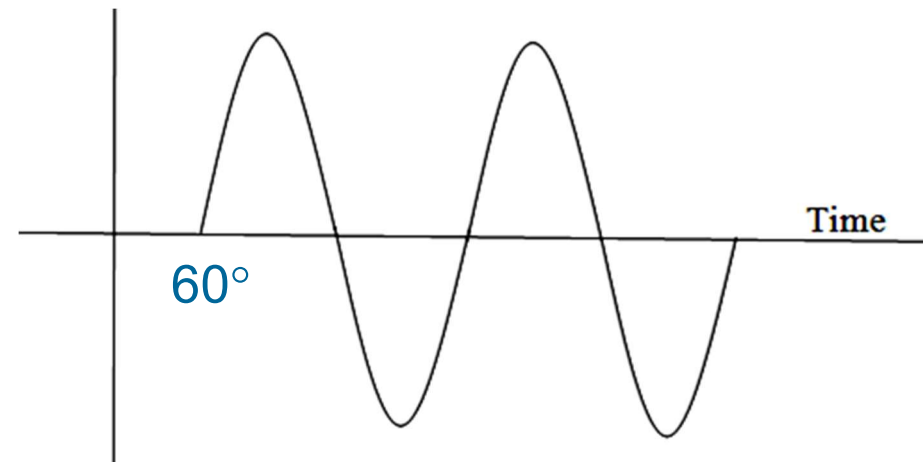
Here the phase is $+30^\circ$
It is leading wrt reference.

$$i = 10 \sin(\omega t + 30^\circ)$$



Here the phase is -60°
It is lagging wrt reference.

$$i = 10 \sin(\omega t - 60^\circ)$$



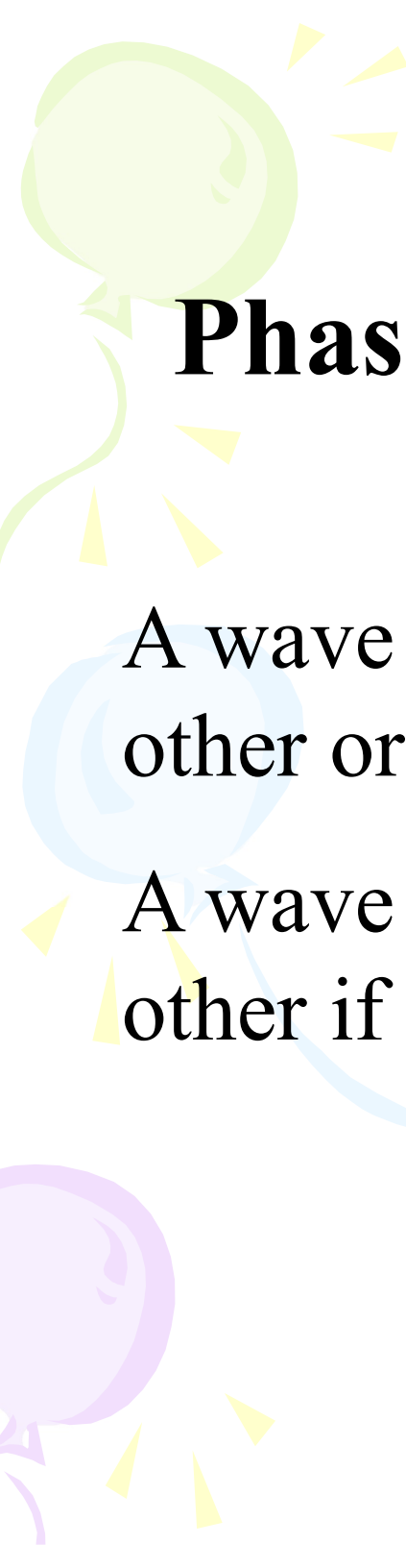
Phase difference

It is the difference of phase angles of the two waves.

$$e = 100 \sin(\omega t + 45^\circ)$$

$$i = 10 \sin(\omega t - 15^\circ)$$

Then phase difference is $= 45^\circ - (-15^\circ) = 60^\circ$



Phase relation (Leading and lagging)

A wave is said to be leading if it occurs before the other or if it reaches the peak value **before** the other

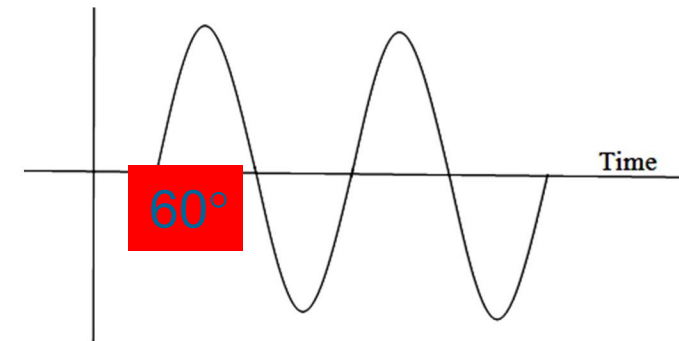
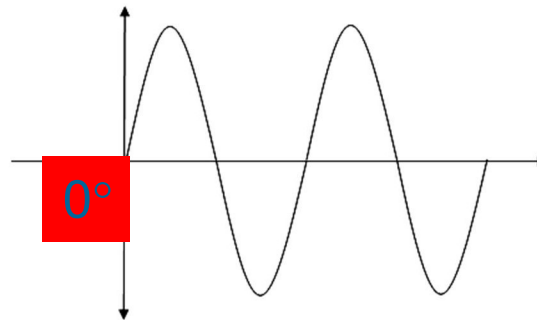
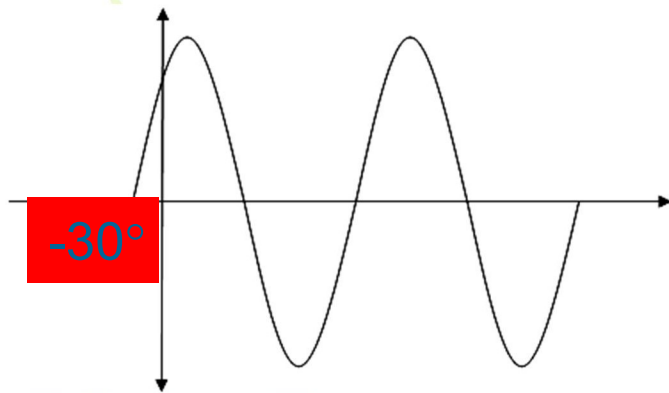
A wave is said to be lagging if it occurs after the other if it reaches the peak value **after** the other

Phase is $+30^\circ$

Phase is
zero.

Phase is -60°

Reference



$$i = 10 \sin(\omega t + 30^\circ)$$

$$i = I_m \sin(\omega t)$$

$$i = 10 \sin(\omega t - 60^\circ)$$

Equation +
Graph left (-)
leading

Equation 0
Graph 0°
Reference

Equation -
Graph right (+)
lagging

Leading wrt reference.

Reference

Lagging wrt reference.

Past (-)

Present (0)

Future (+)

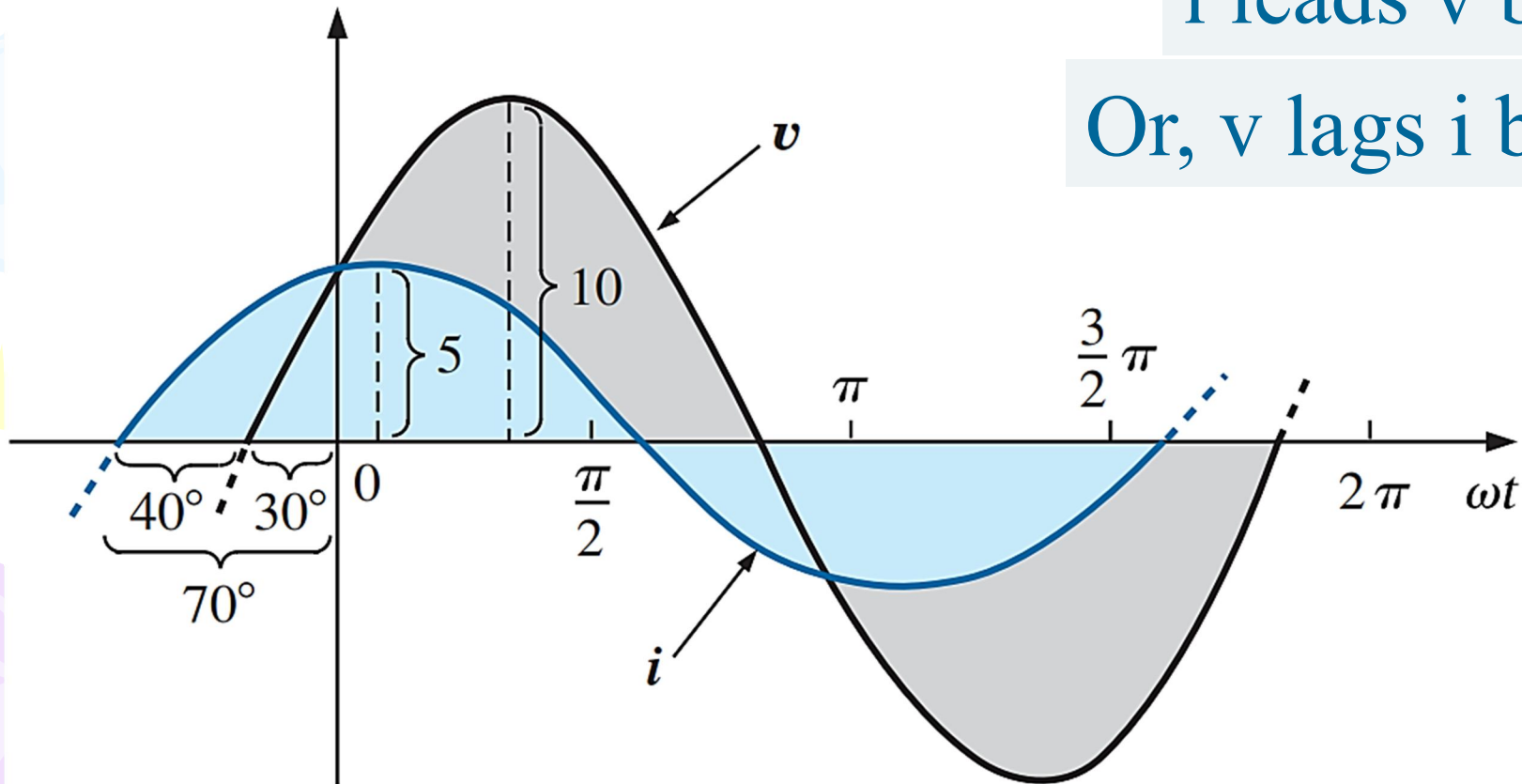
$$v = 10 \sin(\omega t + 30^\circ)$$

$$i = 5 \sin(\omega t + 70^\circ)$$

Phase difference = 40°

i leads v by 40°

Or, v lags i by 40°

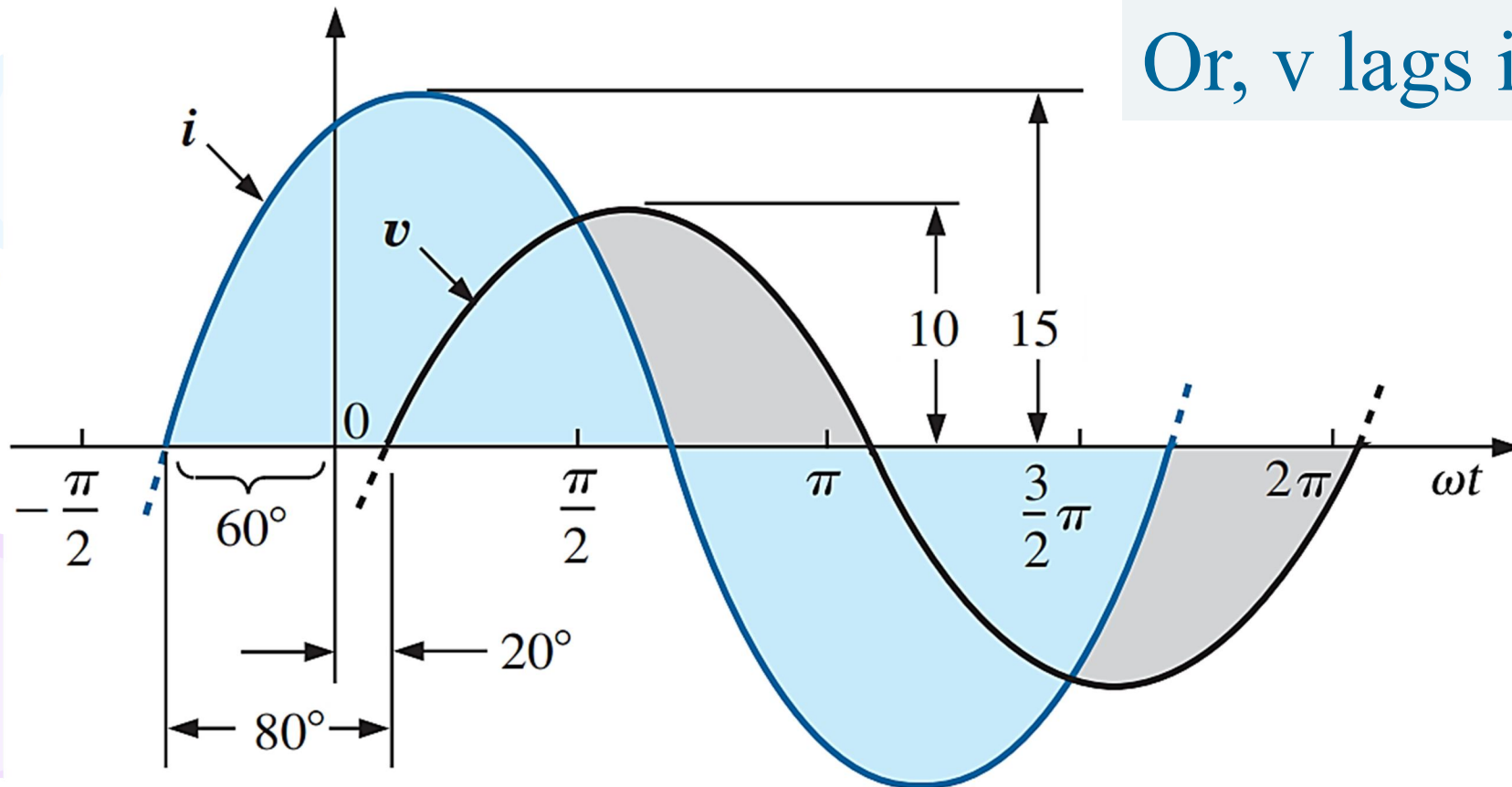


$$i = 15 \sin(\omega t + 60^\circ)$$
$$v = 10 \sin(\omega t - 20^\circ)$$

Phase difference = 80°

i leads v by 80°

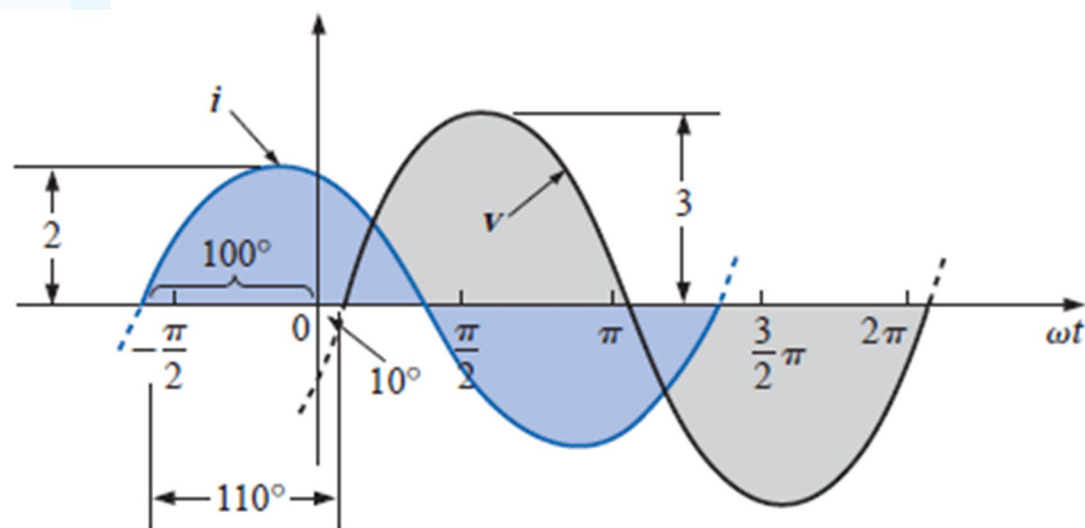
Or, v lags i by 80°



$$i = 2 \cos(\omega t + 10^\circ)$$

$$v = 3 \sin(\omega t - 10^\circ)$$

$$\begin{aligned} i &= 2 \cos(\omega t + 10^\circ) = 2 \sin(\omega t + 10^\circ + 90^\circ) \\ &= 2 \sin(\omega t + 100^\circ) \end{aligned}$$

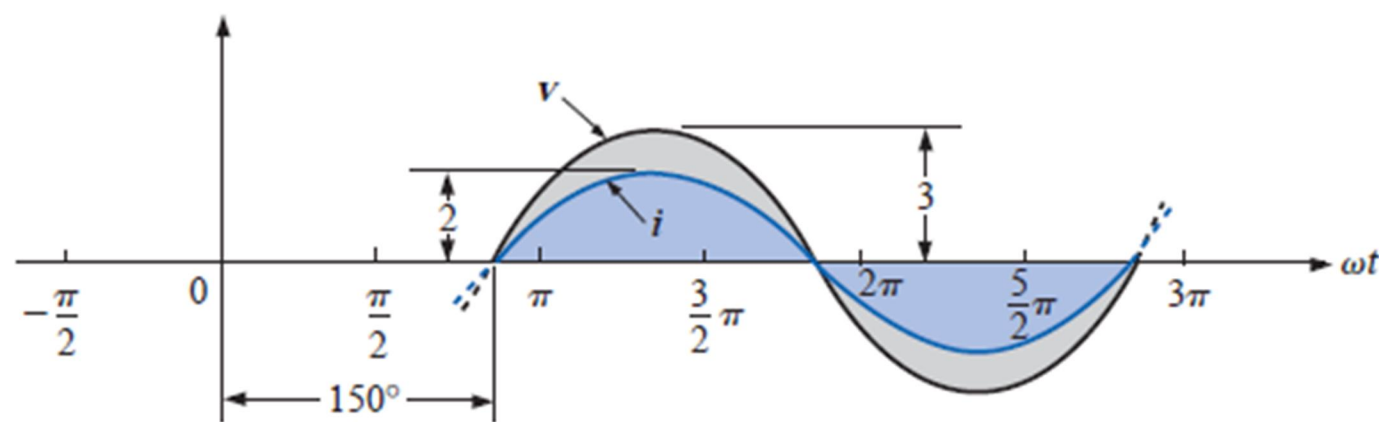


***i* leads *v* by 110° , or *v* lags *i* by 110° .**

$$i = -2 \cos(\omega t - 60^\circ)$$

$$v = 3 \sin(\omega t - 150^\circ)$$

$$\begin{aligned} i &= -2 \cos(\omega t - 60^\circ) = 2 \cos(\omega t - 60^\circ - 180^\circ) \quad \text{By choice} \\ &= 2 \cos(\omega t - 240^\circ) \\ &= 2 \sin(\omega t - 240^\circ + 90^\circ) \\ &= 2 \sin(\omega t - 150^\circ) \end{aligned}$$



v and i are in phase.

$$i = -\sin(\omega t + 30^\circ)$$

$$v = 2 \sin(\omega t + 10^\circ)$$

$$\begin{aligned} -\sin(\omega t + 30^\circ) &= \sin(\omega t + 30^\circ - 180^\circ) \\ &= \sin(\omega t - 150^\circ) \end{aligned}$$

v leads i by 160° , or i lags v by 160° .

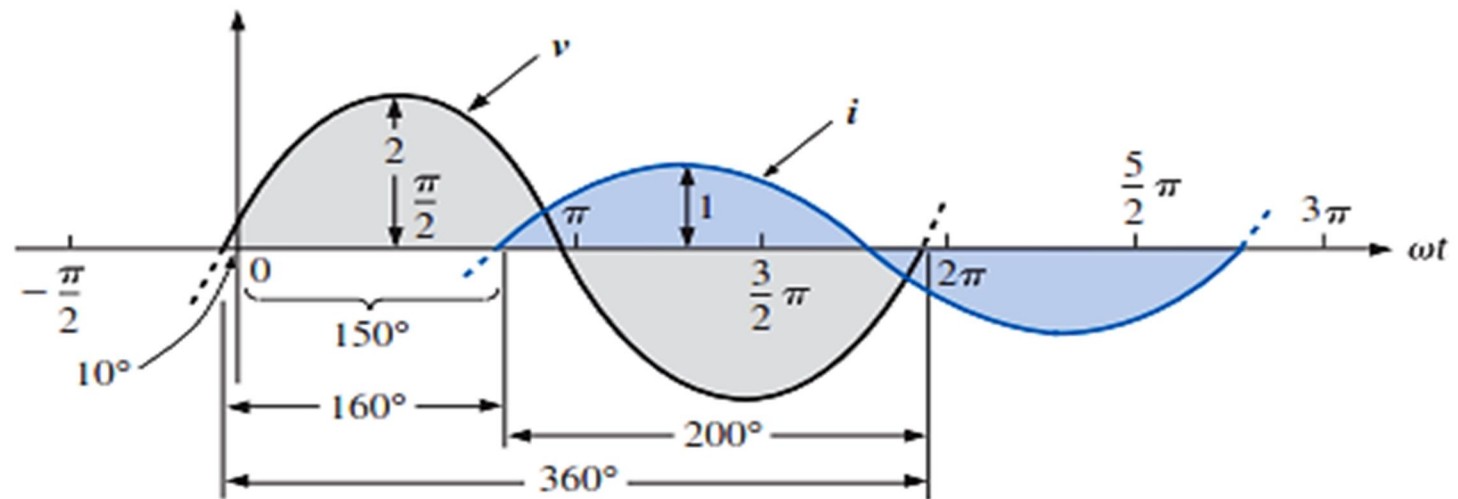


FIG. 13.30

Example 13.12; v leads i by 160° .

Or using

$$\begin{aligned} -\sin(\omega t + 30^\circ) &= \sin(\omega t + 30^\circ + 180^\circ) \\ &= \sin(\omega t + 210^\circ) \end{aligned}$$

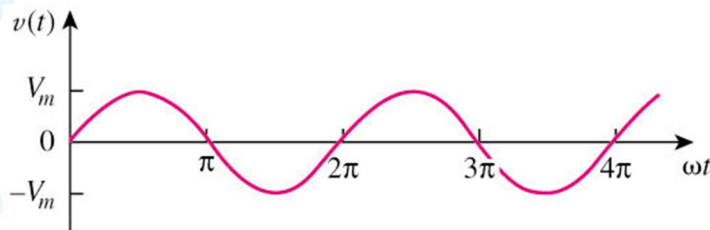
Note

i leads v by 200° , or v lags i by 200° .

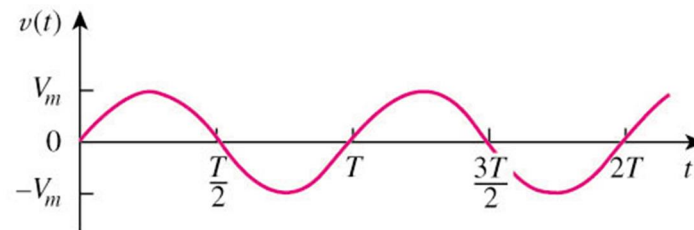
9.2 Sinusoids (1)

- A sinusoid is a signal that has the form of the sine or cosine function.
- A general expression for the sinusoid,

$$v(t) = V_m \sin(\omega t + \phi)$$



(a)



(b)

where

V_m = the **amplitude** of the sinusoid

ω = the angular frequency in radians/s

ϕ = the phase



Sinusoids

Example 1

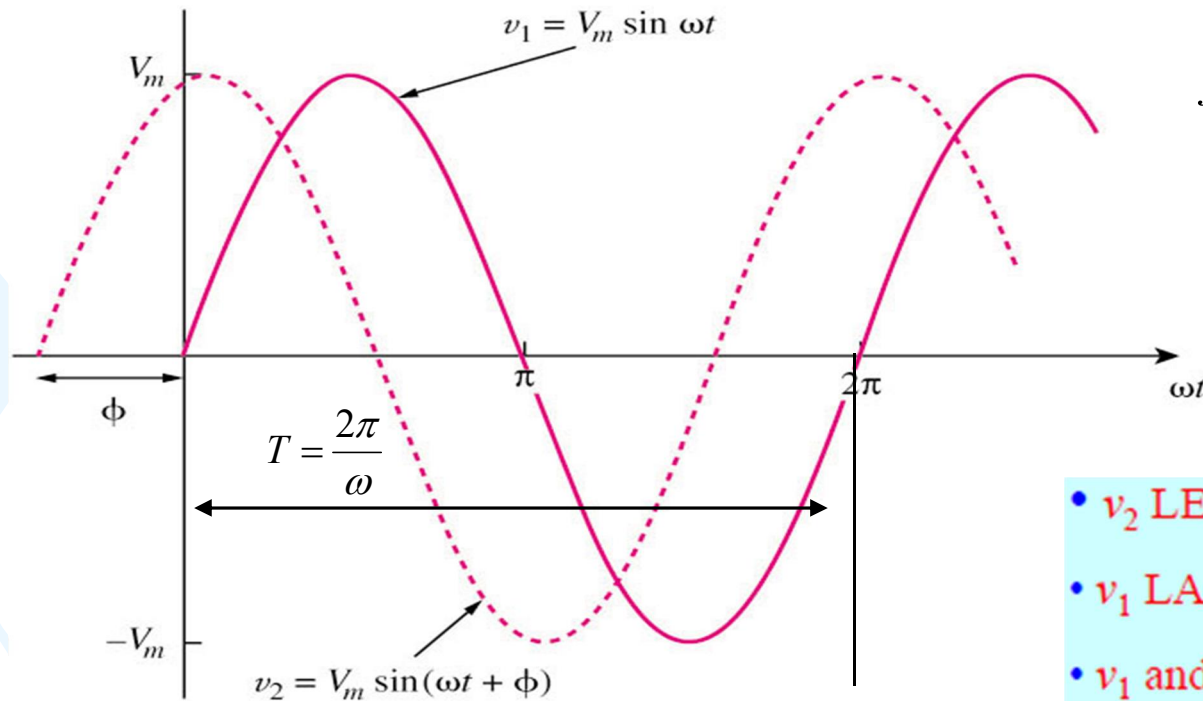
Given a sinusoid, $5 \sin(4\pi t - 60^\circ)$, calculate its amplitude, phase, angular frequency, period, and frequency.

Solution:

Amplitude = 5, phase = -60° , angular frequency = 4π rad/s, Period = 0.5 s, frequency = 2 Hz.

9.2 Phase of Sinusoids

A **periodic function** is one that satisfies $v(t) = v(t + nT)$, for all t and for all integers n .



$$f = \frac{1}{T} \text{ Hz}$$

$$\omega = 2\pi f$$

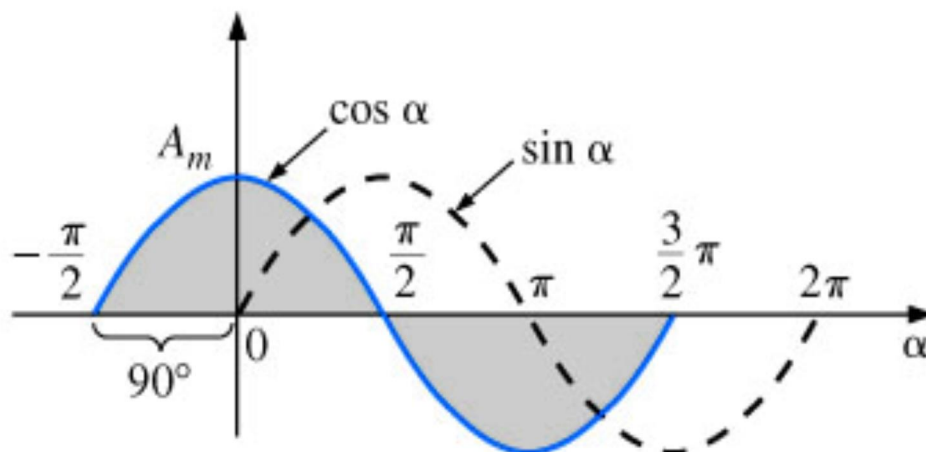
- v_2 LEADS v_1 by phase ϕ .
- v_1 LAGS v_2 by phase ϕ .
- v_1 and v_2 are out of phase.

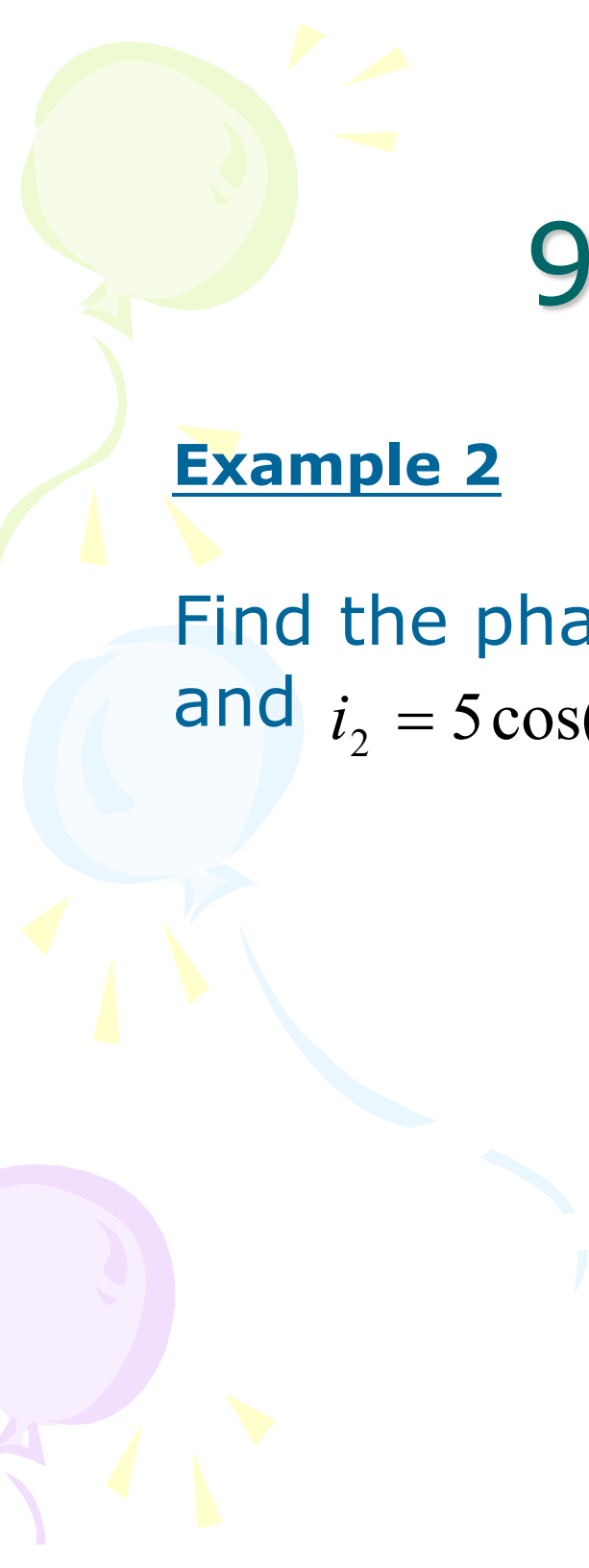
- Only two sinusoidal values with the same frequency can be compared by their amplitude and phase difference.
- If phase difference is zero, they are in phase; if phase difference is not zero, they are out of phase.

Phase of Sinusoids

- The terms *lead* and *lag* are used to indicate the relationship between two sinusoidal waveforms of the *same frequency* plotted on the same set of axes.
- The cosine curve is said to *lead* the sine curve by 90° .
- The sine curve is said to *lag* the cosine curve by 90° .
- 90 is referred to as the phase angle between the two waveforms.
- When determining the phase measurement we first note that each sinusoidal function has the same frequency, permitting the use of either waveform to determine the period.
- Since the full period represents a cycle of 360° , the following ratio can be formed:

$$\theta = \frac{\text{phase shift (no. of div.)}}{T \text{ (no. of div.)}} \times 360^\circ$$





9.2 Sinusoids (4)

Example 2

Find the phase angle between $i_1 = -4 \sin(377t + 25^\circ)$ and $i_2 = 5 \cos(377t - 40^\circ)$, does i_1 lead or lag i_2 ?

9.2 Sinusoids (4)

Example 2

Find the phase angle between $i_1 = -4 \sin(377t + 25^\circ)$ and $i_2 = 5 \cos(377t - 40^\circ)$, does i_1 lead or lag i_2 ?

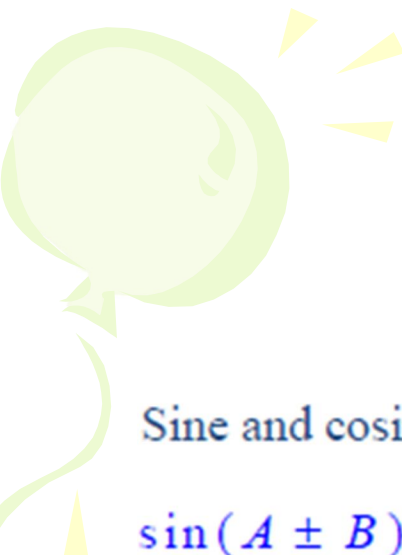
Solution:

Since $\sin(\omega t + 90^\circ) = \cos \omega t$

$$i_2 = 5 \sin(377t - 40^\circ + 90^\circ) = 5 \sin(377t + 50^\circ)$$

$$i_1 = -4 \sin(377t + 25^\circ) = 4 \sin(377t + 180^\circ + 25^\circ) = 4 \sin(377t + 205^\circ)$$

therefore, i_1 leads i_2 155° .



Sine and cosine form conversions.

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$


$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

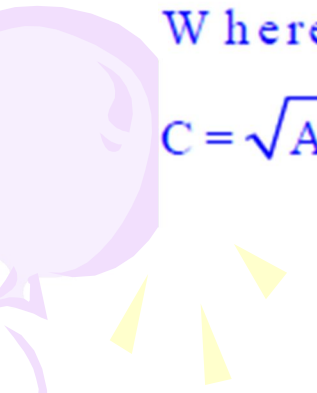
$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

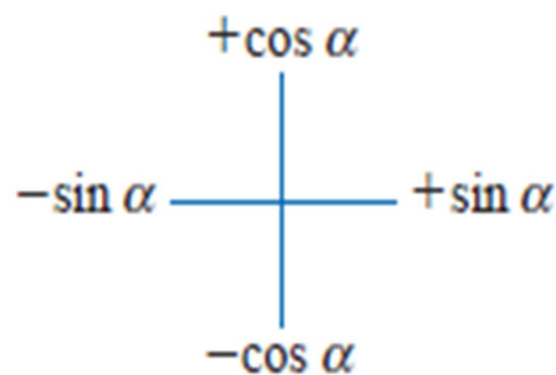
$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

$$A \cos \omega t + B \sin \omega t = C \cos(\omega t - \theta)$$

Where

$$C = \sqrt{A^2 + B^2} \quad \text{and} \quad \theta = \tan^{-1} \frac{B}{A}$$




EXERCISE

➤ Voltage and current are out of phase by 40° , and voltage lags. Using current as the reference, sketch the phasor diagram and the corresponding waveforms.

Voltage and current are out of phase by 40° , and voltage lags. Using current as the reference, sketch the phasor diagram and the corresponding waveforms.

Solution Since current is the reference, place its phasor in the 0° position and the voltage phasor at -40° . Figure 15-42 shows the phasors and corresponding waveforms.

